



Digital Design

Chapter 4: Datapath Components

Slides to accompany the textbook *Digital Design*, First Edition,
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<http://www.ddvahid.com>

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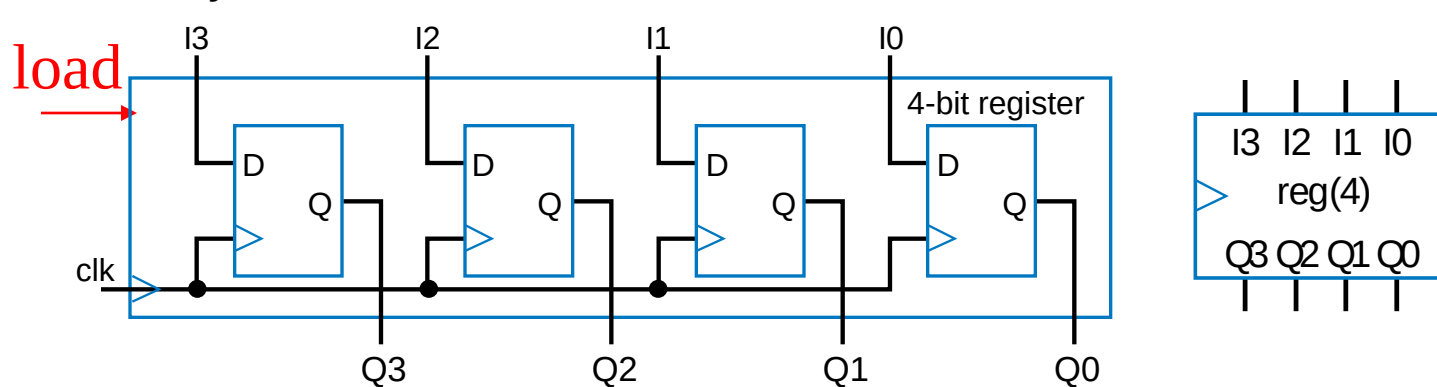
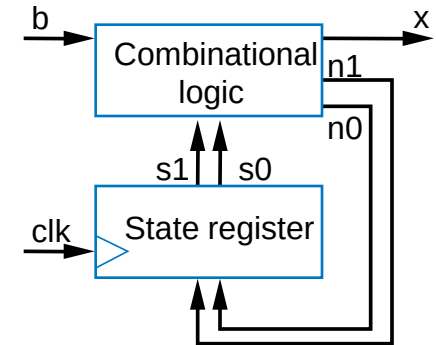
Introduction

- Chapters 2 & 3: Introduced increasingly complex digital building blocks
 - Gates, multiplexors, decoders, basic registers, and controllers
- Controllers good for systems with control inputs/outputs
 - **Control** input: Single bit (or just a few), representing environment event or state
 - e.g., 1 bit representing button pressed
 - **Data** input: Multiple bits collectively representing single entity
 - e.g., 7 bits representing temperature in binary
- Need building blocks for data sis
 - **Datapath components**, aka register-transfer-level (RTL) components, store/transform data
 - Put datapath components together to form a **datapath**
- This chapter introduces numerous datapath components, and simple datapaths
 - Next chapter will combine controllers and datapaths into “processors”



Registers

- Can store data, very common in datapaths
- Basic register of Ch 3: Loaded every cycle
 - Useful for implementing FSM -- stores encoded state
 - For other uses, may want to load only on certain cycles



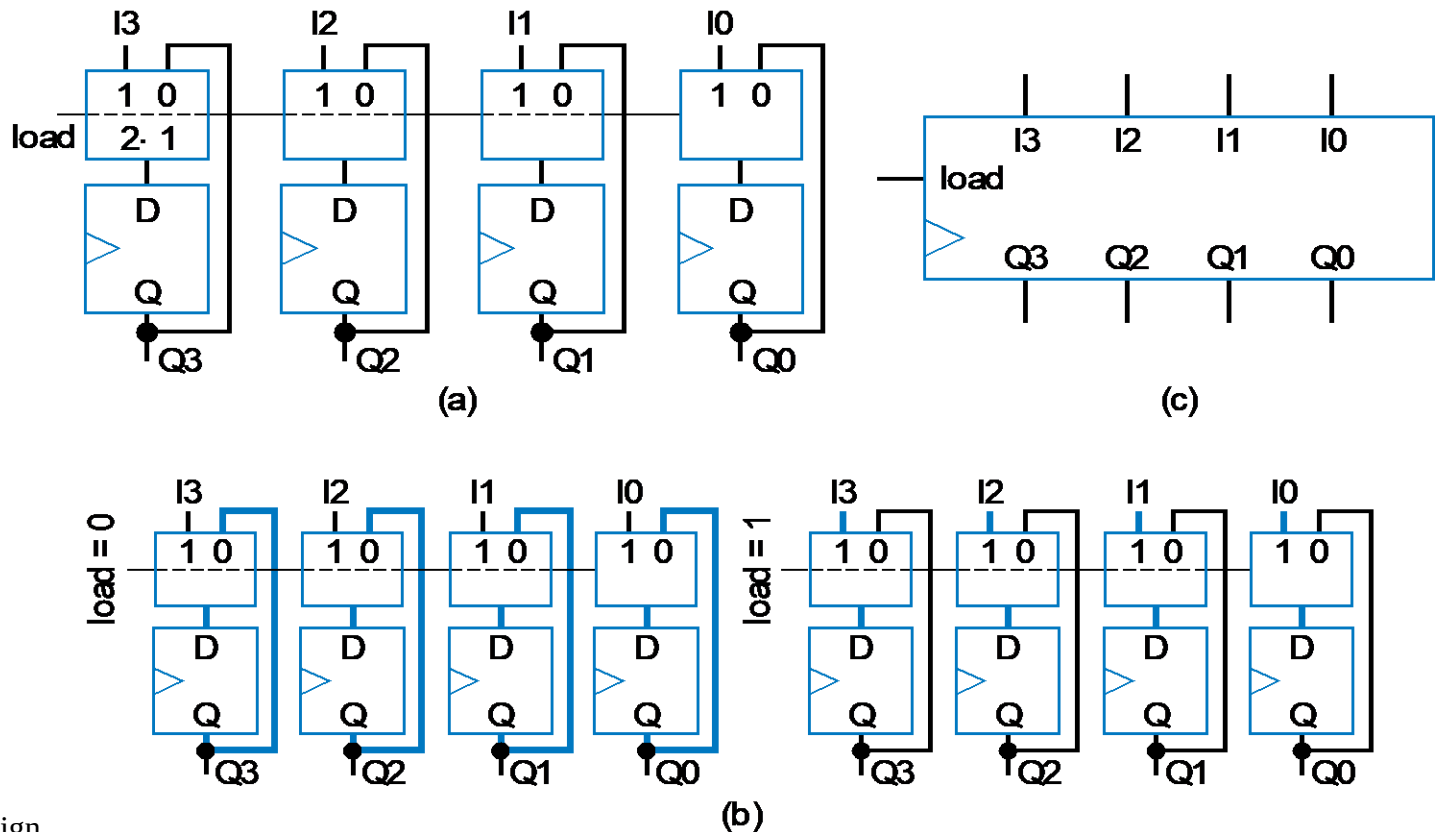
Basic register loads on every clock cycle

How extend to only load on certain cycles?



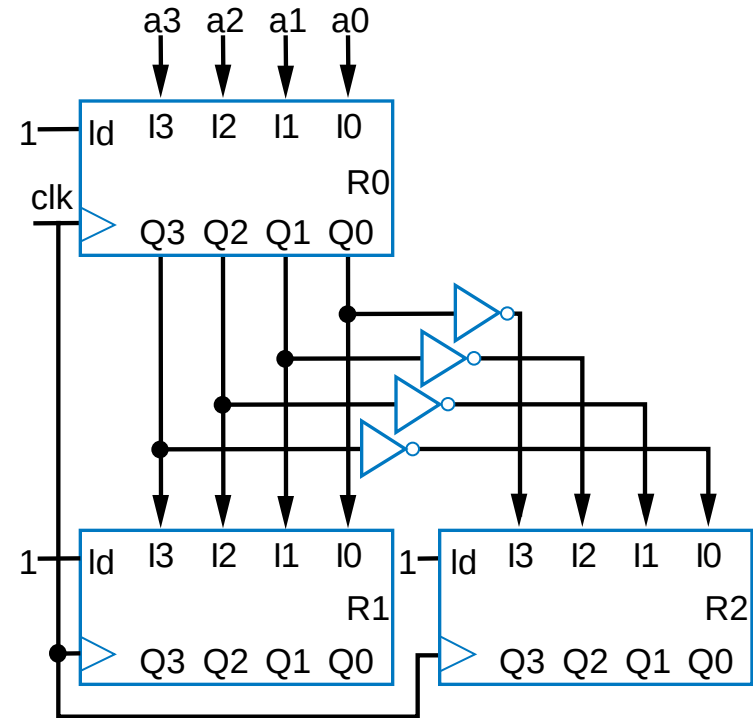
Register with Parallel Load

- Add 2x1 mux to front of each flip-flop
- Register's load input selects mux input to pass
 - Either existing flip-flop value, or new value to load

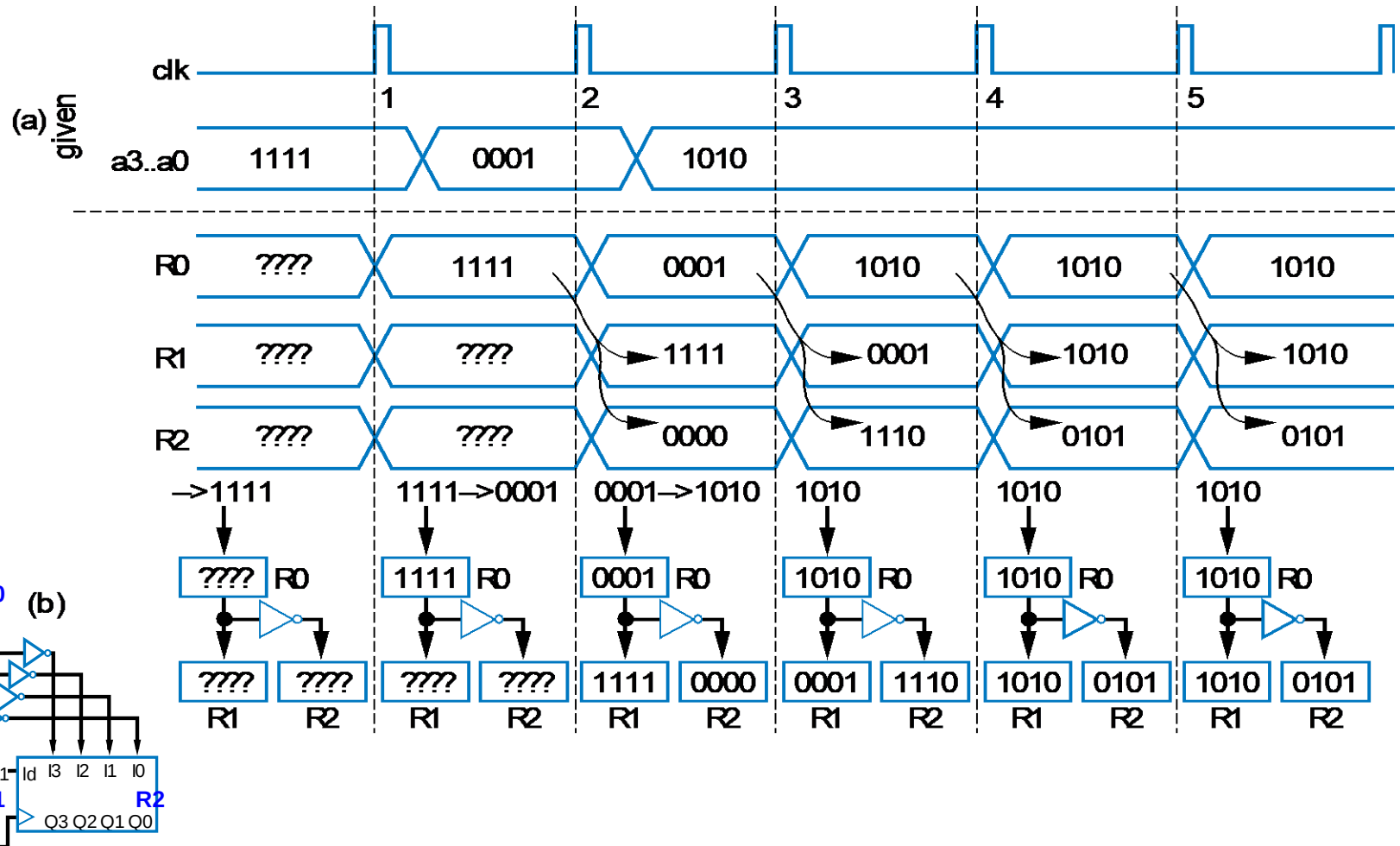


Basic Example Using Registers

- This example will show how registers load simultaneously on clock cycles
 - Notice that all load inputs set to 1 in this example -- just for demonstration purposes

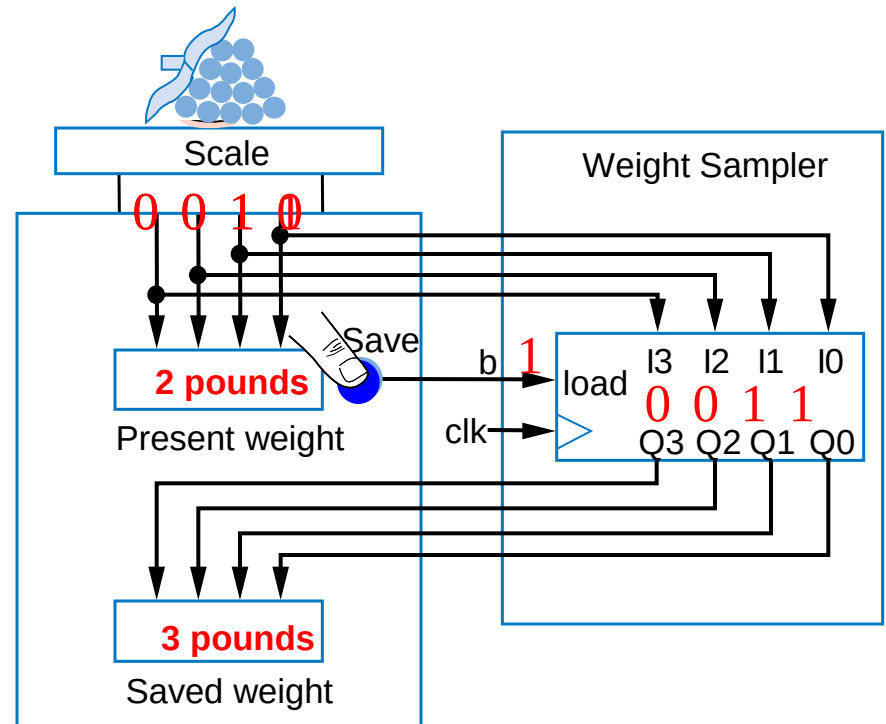


Basic Example Using Registers



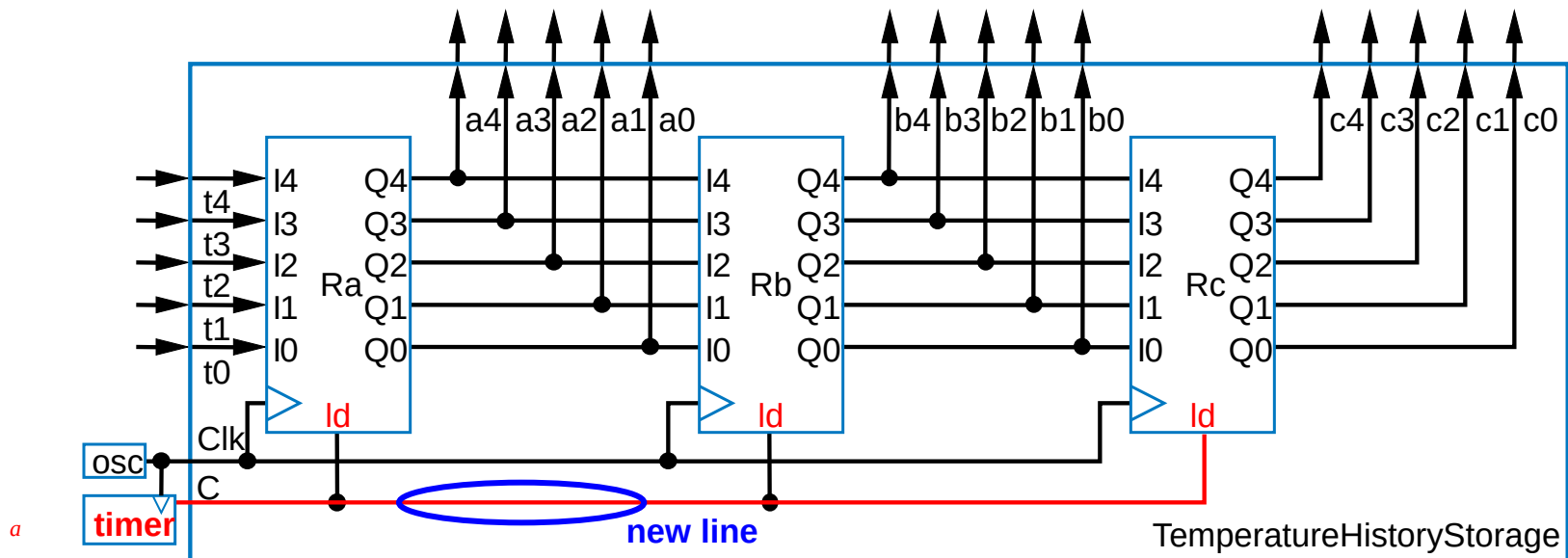
Register Example using the Load Input: Weight Sampler

- Scale has two displays
 - Present weight
 - Saved weight
 - Useful to compare present item with previous item
- Use register to store weight
 - Pressing button causes present weight to be stored in register
 - Register contents always displayed as “Saved weight,” even when new present weight appears

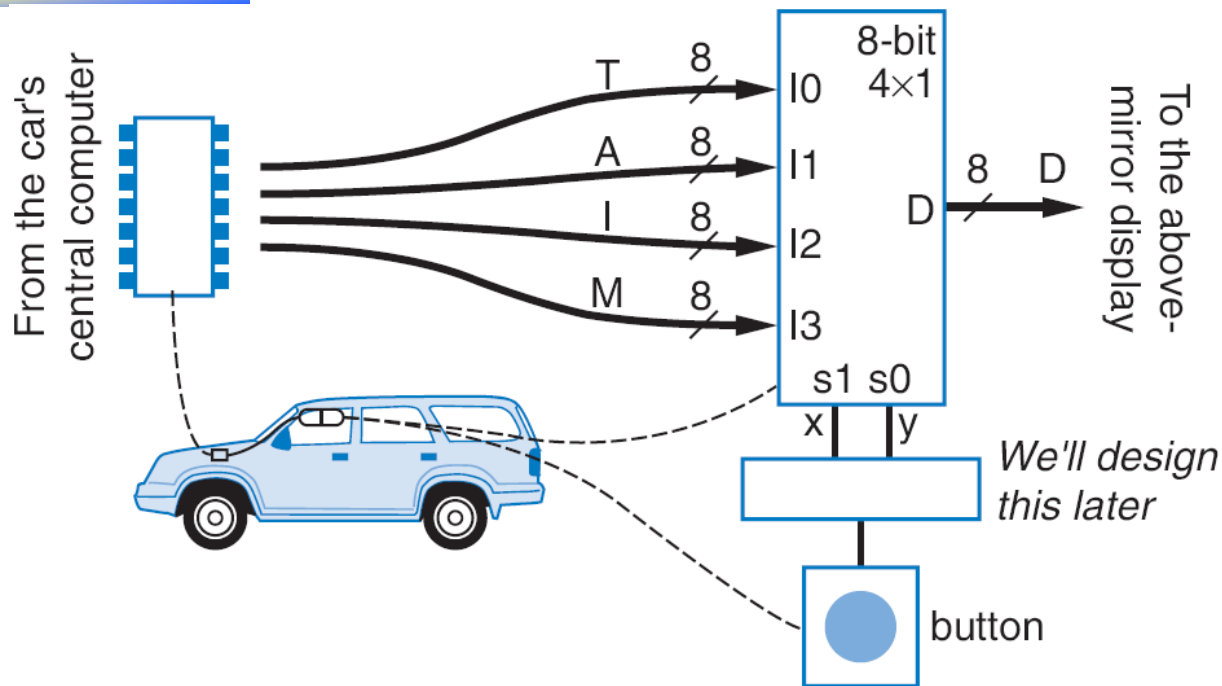


Register Example: Temperature History Display

- Recall Chpt 3 example
 - Timer pulse every hour
 - Previously used as clock. Better design only connects oscillator to clock inputs -- use registers with load input, connect to timer pulse.



N-bit Mux Example

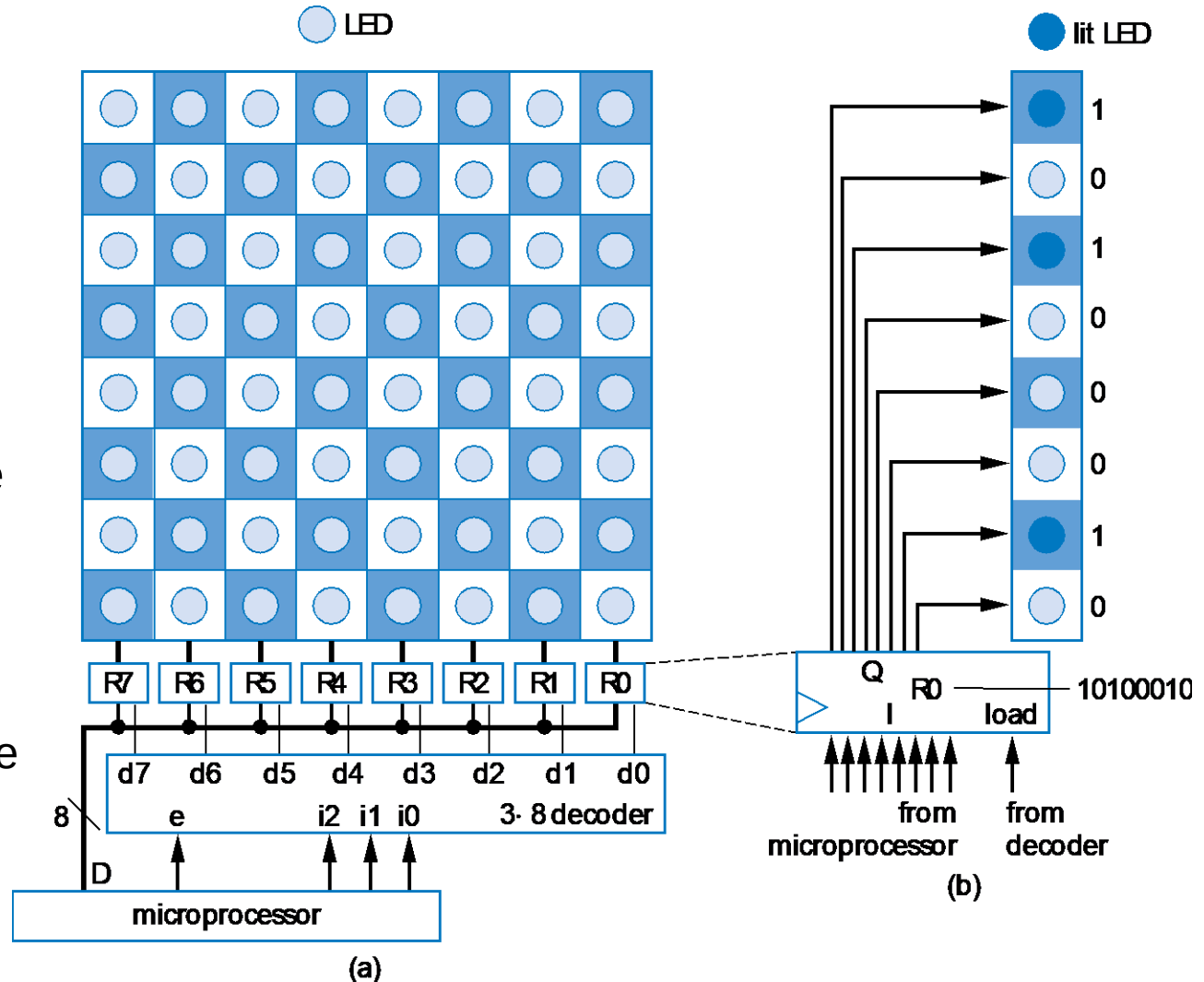


- Four possible display items
 - Temperature (T), Average miles-per-gallon (A), Instantaneous mpg (I), and Miles remaining (M) -- each is 8-bits wide
 - Choose which to display using two inputs x and y
 - Use 8-bit 4x1 mux

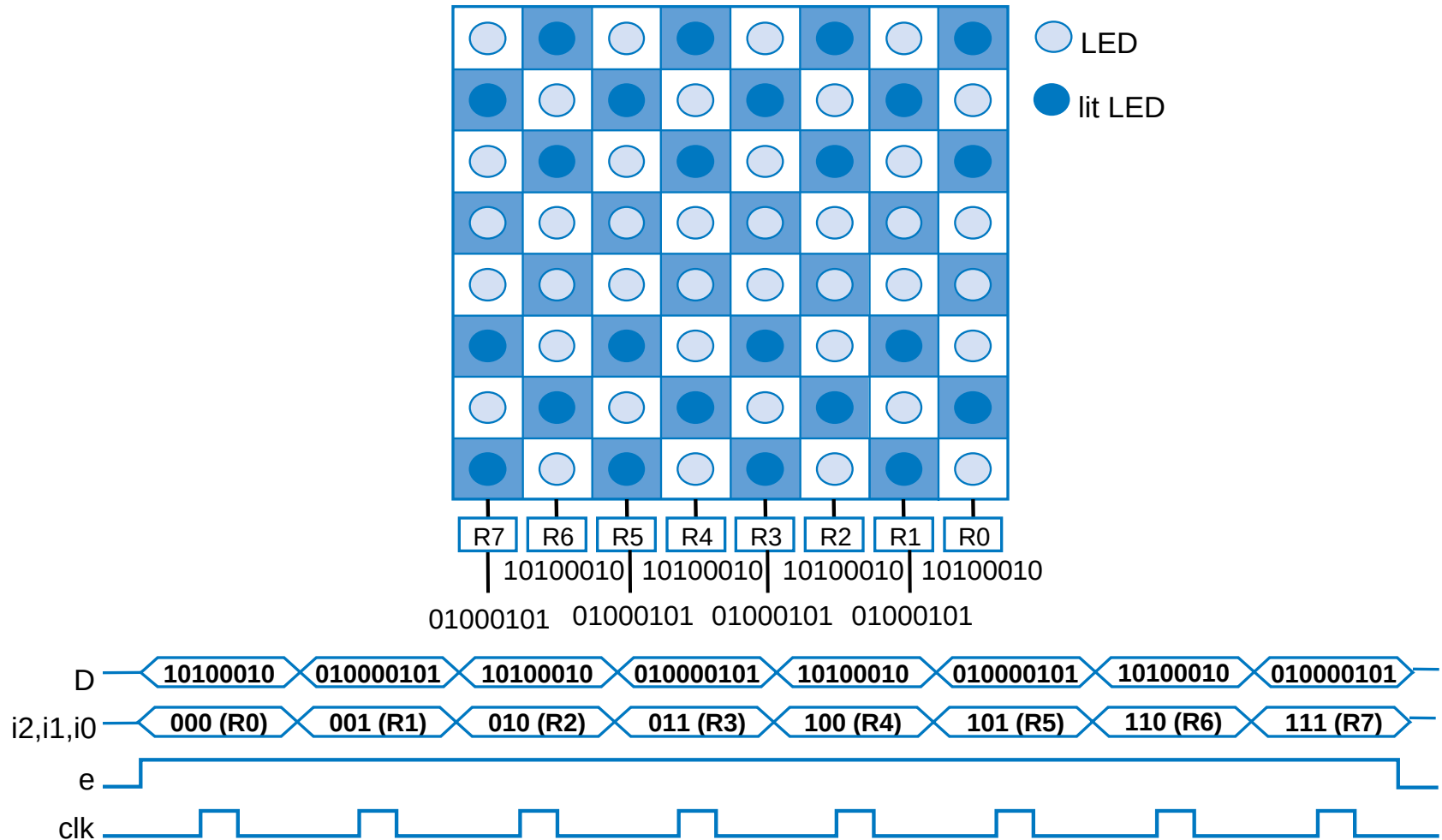


Register Example: Computerized Checkerboard

- Each register holds values for one column of lights
 - 1 lights light
- Microprocessor loads one register at a time
 - Occurs fast enough that user sees entire board change at once

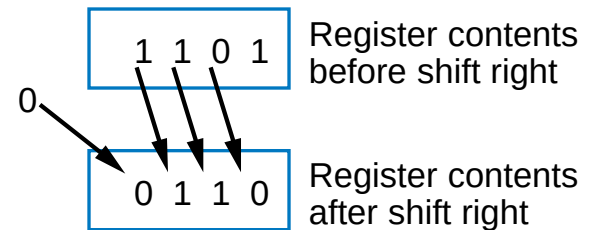


Register Example: Computerized Checkerboard



Shift Register

- Shift right
 - Move each bit one position right
 - Shift in 0 to leftmost bit

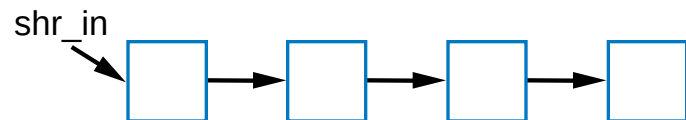


Q: Do four right shifts on 1001, showing value after each shift

A:

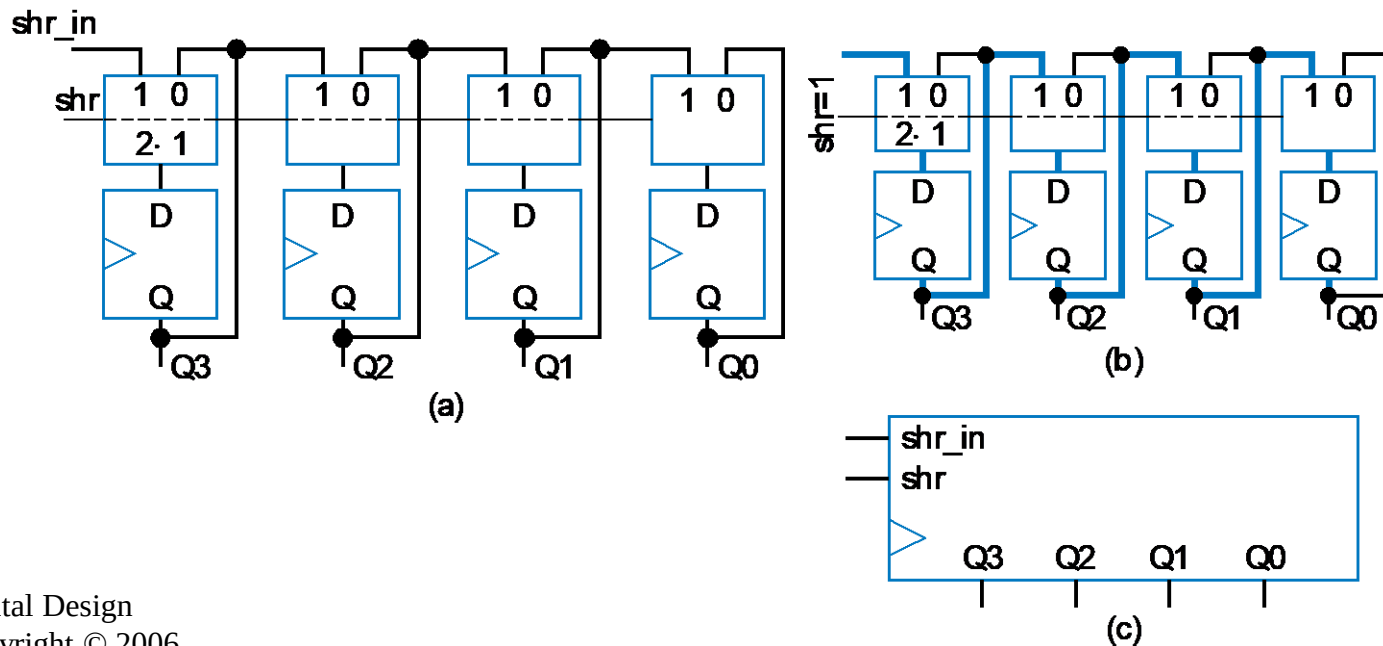
1001	(original)
0100	
0010	
0001	
0000	

- Implementation: Connect flip-flop output to next flip-flop's input



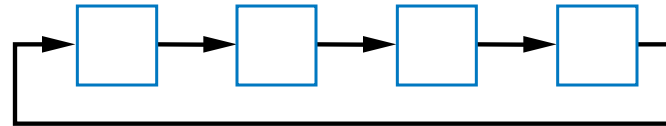
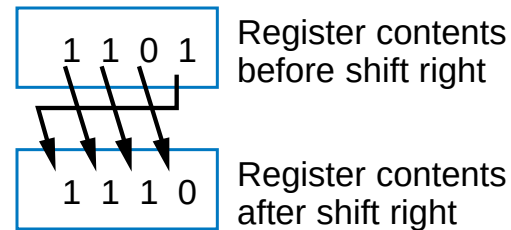
Shift Register

- To allow register to either shift or retain, use 2x1 muxes
 - shr: 0 means retain, 1 shift
 - shr_in: value to shift in
 - May be 0, or 1
- Note: Can easily design shift register that shifts left instead



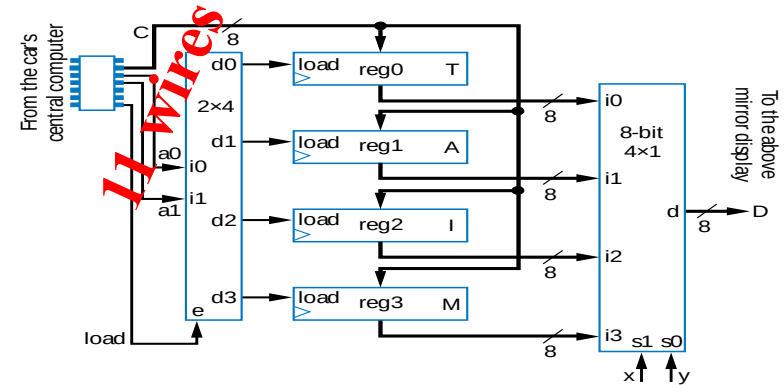
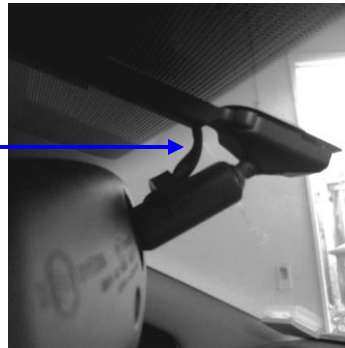
Rotate Register

- Rotate right: Like shift right, but leftmost bit comes from rightmost bit

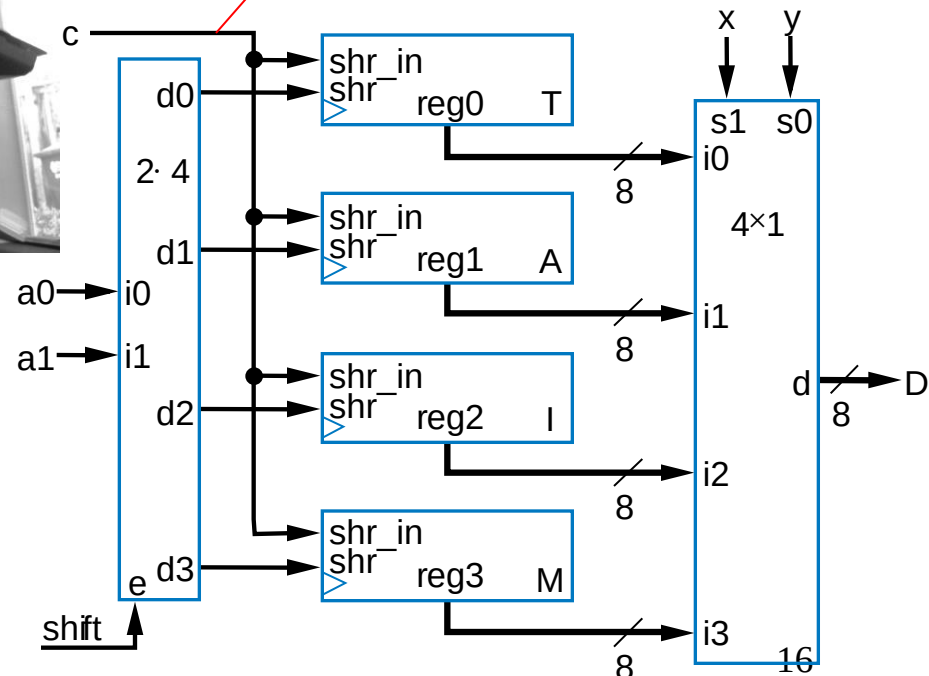


Shift Register Example: Above-Mirror Display

- Earlier example: 8 + 2 + 1 = 11 wires from car's computer to above-mirror display's four registers
 - Better than 32 wires, but 11 still a lot -- want fewer for smaller wire bundles
- Use shift registers
 - Wires: 1 + 2 + 1 = 4
 - Computer sends one value at a time, one bit per clock cycle



Note: this line is 1 bit, rather than 8 bits like before

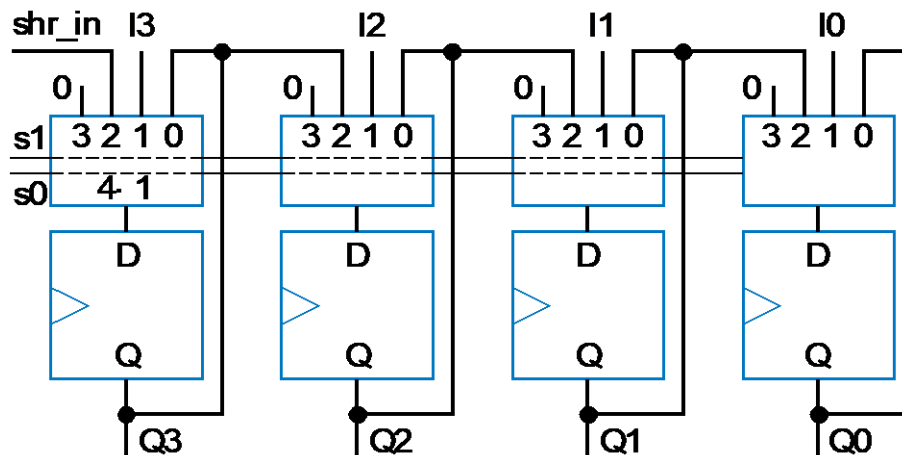


Multifunction Registers

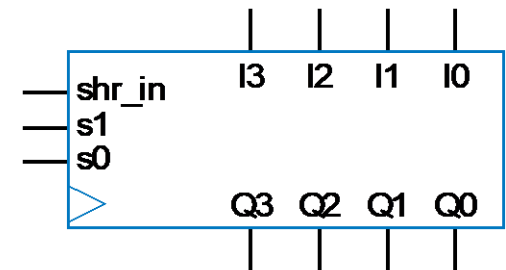
- Many registers have multiple functions
 - Load, shift, clear (load all 0s)
 - And retain present value, of course
- Easily designed using muxes
 - Just connect each mux input to achieve desired function

Functions:

s1	s0	Operation
0	0	Maintain present value
0	1	Parallel load
1	0	Shift right
1	1	(unused - let's load 0s)



(a)

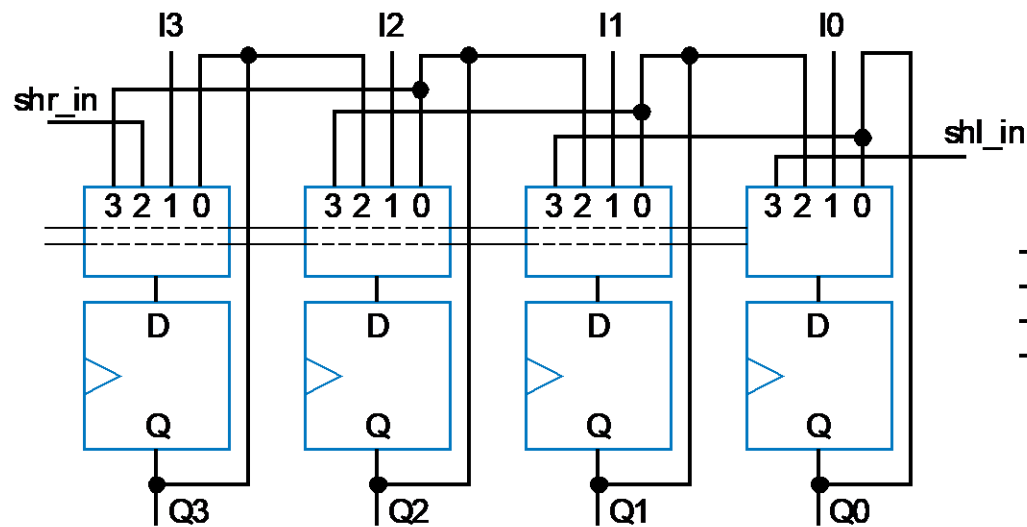


(b)

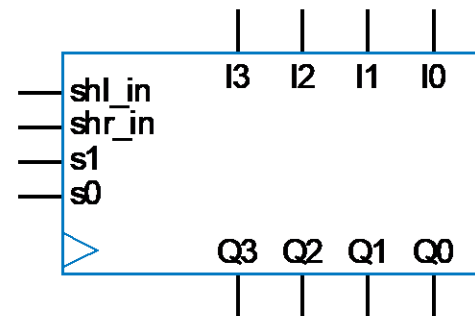


Multifunction Registers

s1	s0	Operation
0	0	Maintain present value
0	1	Parallel load
1	0	Shift right
1	1	Shift left



(a)



(b)

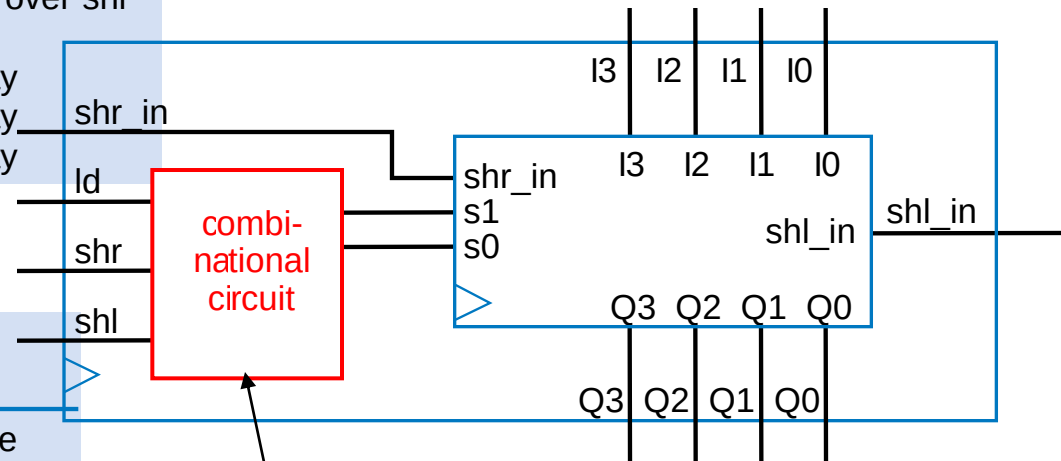


Multifunction Registers with Separate Control Inputs

ld	shr	shl	Operation
0	0	0	Maintain present value
0	0	1	Shift left
0	1	0	Shift right
0	1	1	Shift right – shr has priority over shl
1	0	0	Parallel load
1	0	1	Parallel load – ld has priority
1	1	0	Parallel load – ld has priority
1	1	1	Parallel load – ld has priority

Truth table for combinational circuit

Inputs			Outputs		Note Operation
ld	shr	shl	s1	s0	
0	0	0	0	0	Maintain value
0	0	1	1	1	Shift left
0	1	0	1	0	Shift right
0	1	1	1	0	Shift right
1	0	0	0	1	Parallel load
1	0	1	0	1	Parallel load
1	1	0	0	1	Parallel load
1	1	1	0	1	Parallel load



$$s1 = ld' * shr' * shl + ld' * shr * shl' + ld' * shr * shl$$

$$s0 = ld' * shr' * shl + ld$$



Register Operation Table

- Register operations typically shown using compact version of table
 - X means same operation whether value is 0 or 1
 - One X expands to two rows
 - Two Xs expand to four rows
 - Put highest priority control input on left to make reduced table simple

Inputs			Outputs		Note Operation
ld	shr	shl	s1	s0	
0	0	0	0	0	Maintain value
0	0	1	1	1	Shift left
0	1	0	1	0	Shift right
0	1	1	1	0	Shift right
1	0	0	0	1	Parallel load
1	0	1	0	1	Parallel load
1	1	0	0	1	Parallel load
1	1	1	0	1	Parallel load

ld	shr	shl	Operation
ld	shr	shl	
0	0	0	Maintain value
0	0	1	Shift left
0	1	X	Shift right
1	X	X	Parallel load



Register Design Process

- Can design register with desired operations using simple four-step process

TABLE 4.1 Four-step process for designing a multifunction register.

	Step	Description
1.	<i>Determine mux size</i>	Count the number of operations (don't forget the maintain present value operation!) and add in front of each flip-flop a mux with at least that number of inputs.
2.	<i>Create mux operation table</i>	Create an operation table defining the desired operation for each possible value of the mux select lines.
3.	<i>Connect mux inputs</i>	For each operation, connect the corresponding mux data input to the appropriate external input or flip-flop output (possibly passing through some logic) to achieve the desired operation.
4.	<i>Map control lines</i>	Create a truth table that maps external control lines to the internal mux select lines, with appropriate priorities, and then design the logic to achieve that mapping



Register Design Example

- Desired register operations
 - Load, shift left, synchronous clear, synchronous set

Step 1: Determine mux size

5 operations: above, plus maintain present value (don't forget this one!)

--> **Use 8x1 mux**

Step 2: Create mux operation table

Step 3: Connect mux inputs

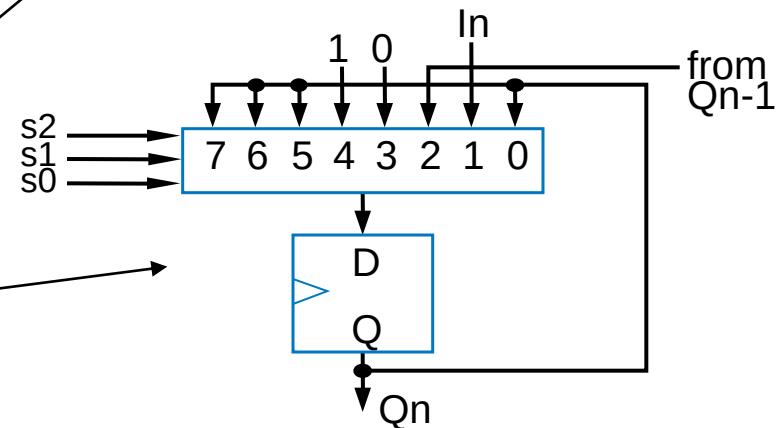
Step 4: Map control lines

$$s2 = \text{clr}' * \text{set}$$

$$s1 = \text{clr}' * \text{set}' * \text{ld}' * \text{shl} + \text{clr}$$

$$s0 = \text{clr}' * \text{set}' * \text{ld} + \text{clr}$$

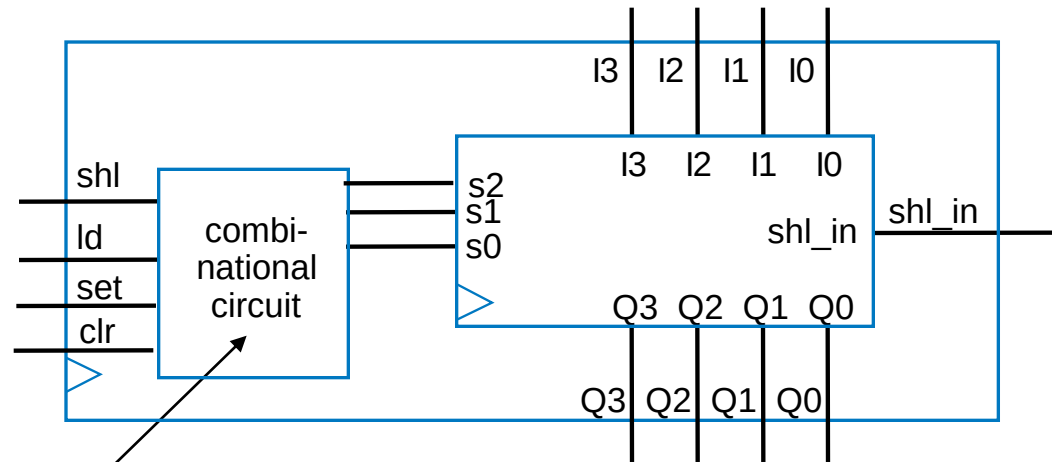
s2	s1	s0	Operation
0	0	0	Maintain present value
0	0	1	Parallel load
0	1	0	Shift left
0	1	1	Synchronous clear
1	0	0	Synchronous set
1	0	1	Maintain present value
1	1	0	Maintain present value
1	1	1	Maintain present value



Inputs				Outputs			Operation
clr	set	ld	shl	s2	s1	s0	
0	0	0	0	0	0	0	Maintain present value
0	0	0	1	0	1	0	Shift left
0	0	1	X	0	0	1	Parallel load
0	1	X	X	1	0	0	Set to all 1s
1	X	X	X	0	1	1	Clear to all 0s



Register Design Example



Step 4: Map control lines

$$s2 = \text{clr}' * \text{set}$$

$$s1 = \text{clr}' * \text{set}' * \text{ld}' * \text{shl} + \text{clr}$$

$$s0 = \text{clr}' * \text{set}' * \text{ld} + \text{clr}$$

Inputs				Outputs			Operation
clr	set	ld	shl	s2	s1	s0	
0	0	0	0	0	0	0	Maintain present value
0	0	0	1	0	1	0	Shift left
0	0	1	X	0	0	1	Parallel load
0	1	X	X	1	0	0	Set to all 1s
1	X	X	X	0	1	1	Clear to all 0s



Adders

- Adds two N-bit binary numbers
 - 2-bit adder: adds two 2-bit numbers, outputs 3-bit result
 - e.g., $01 + 11 = 100$ ($1 + 3 = 4$)
- Can design using combinational design process of Ch 2, but doesn't work well for reasonable-size N
 - Why not?

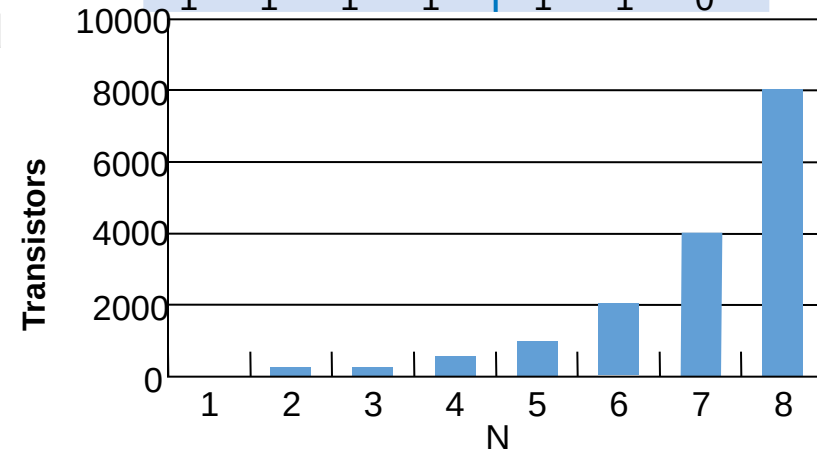
Inputs				Outputs		
a1	a0	b1	b0	c	s1	s0
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0



Why Adders Aren't Built Using Standard Combinational Design Process

- Truth table too big
 - 2-bit adder's truth table shown
 - Has $2^{(2+2)} = 16$ rows
 - 8-bit adder: $2^{(8+8)} = 65,536$ rows
 - 16-bit adder: $2^{(16+16)} = \sim 4$ billion rows
 - 32-bit adder: ...
- Big truth table with numerous 1s/0s yields big logic
 - Plot shows number of transistors for N-bit adders, using state-of-the-art automated combinational design tool

Inputs				Outputs		
a1	a0	b1	b0	c	s1	s0
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	1
1	0	1	1	1	0	0
1	1	0	0	0	1	0
1	1	0	1	0	1	1
1	1	1	0	1	0	0
1	1	1	1	1	0	1



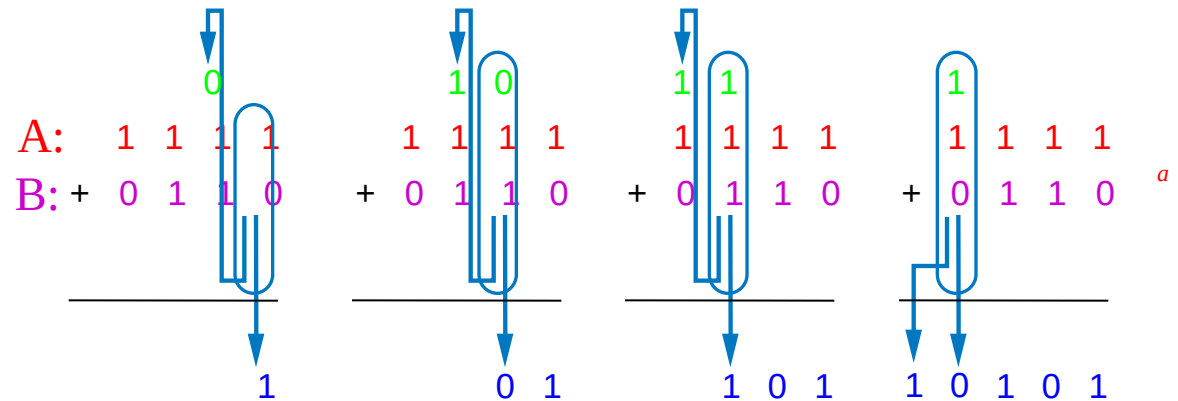
Q: Predict number of transistors for 16-bit adder

A: 1000 transistors for N=5, doubles for each increase of N. So transistors = $1000 \cdot 2^{(N-5)}$. Thus, for N=16, transistors = $1000 \cdot 2^{(16-5)} = 1000 \cdot 2048 = 2,048,000$. Way too many!



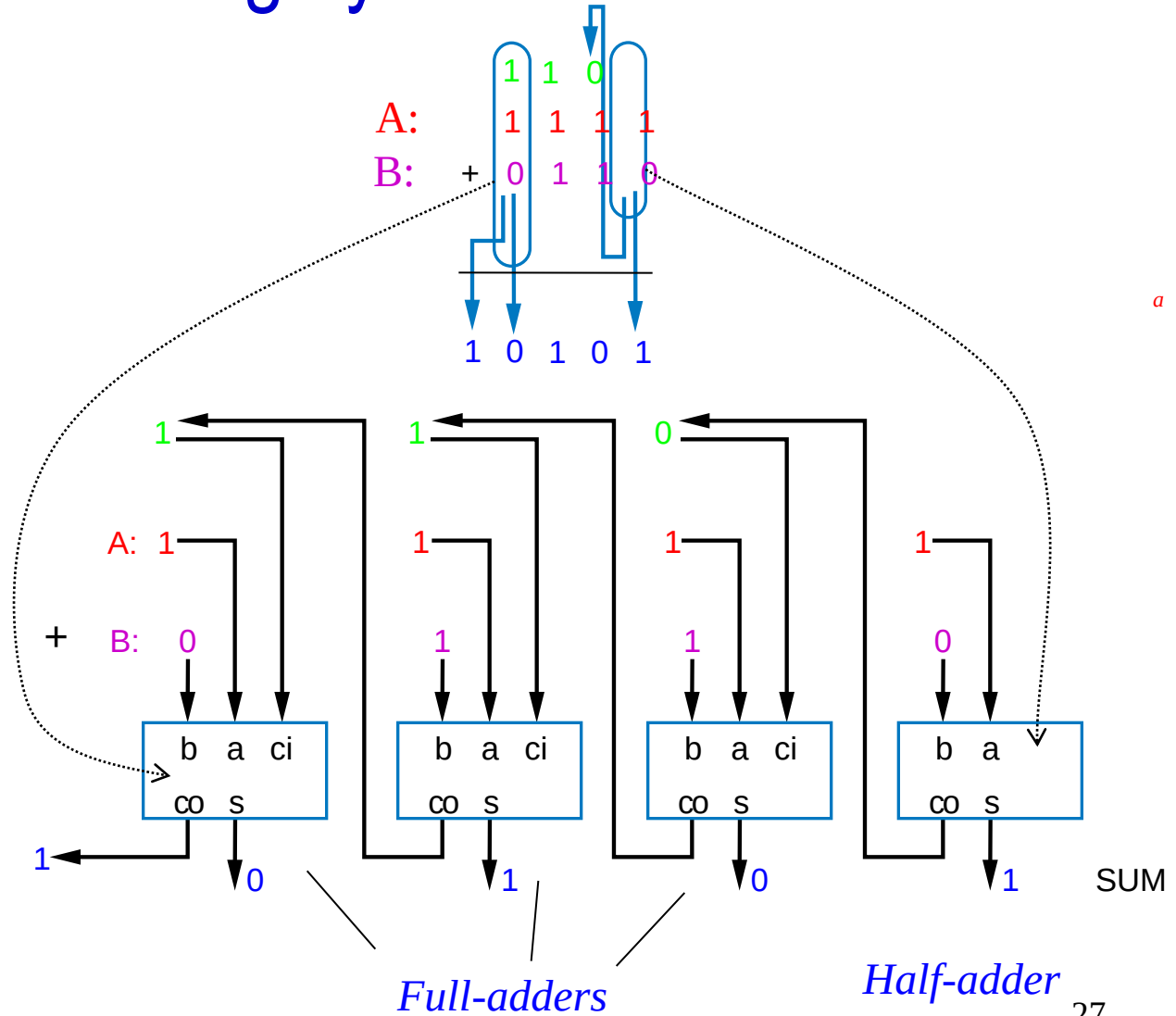
Alternative Method to Design an Adder: Imitate Adding by Hand

- Alternative adder design: mimic how people do addition by hand
- One column at a time
 - Compute sum, add carry to next column



Alternative Method to Design an Adder: Imitate Adding by Hand

- Create component for each column
 - Adds that column's bits, generates sum and carry bits



Half-Adder

- **Half-adder**: Adds 2 bits, generates sum and carry
- Design using combinational design process from Ch 2

Step 1: Capture the function

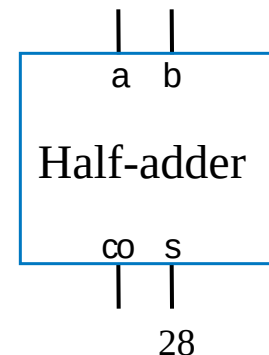
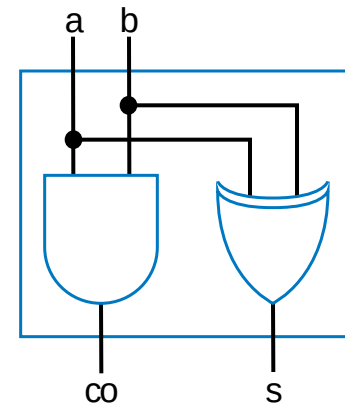
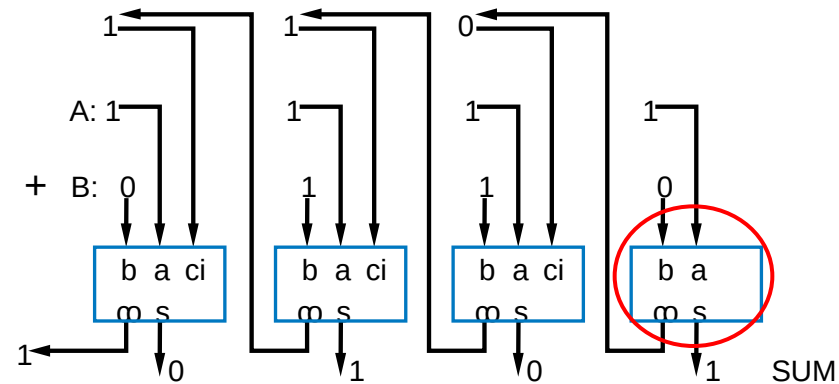
Inputs		Outputs	
a	b	co	s
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Step 2: Convert to equations

$$co = ab$$

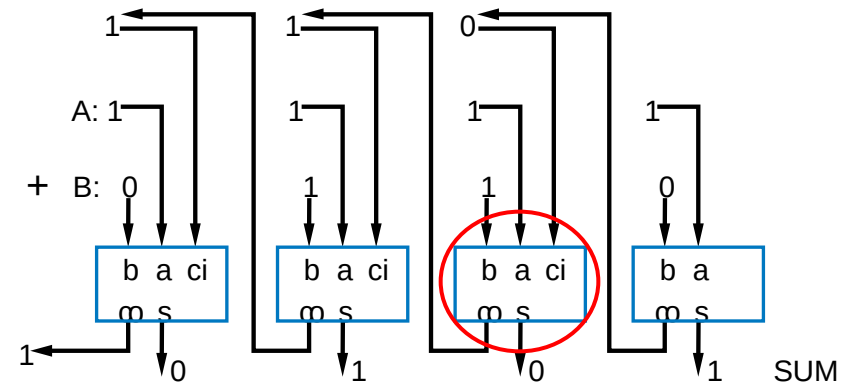
$$s = a'b + ab' \text{ (same as } s = a \text{ xor } b)$$

Step 3: Create the circuit



Full-Adder

- **Full-adder:** Adds 3 bits, generates sum and carry
- Design using combinational design process from Ch 2



Step 1: Capture the function

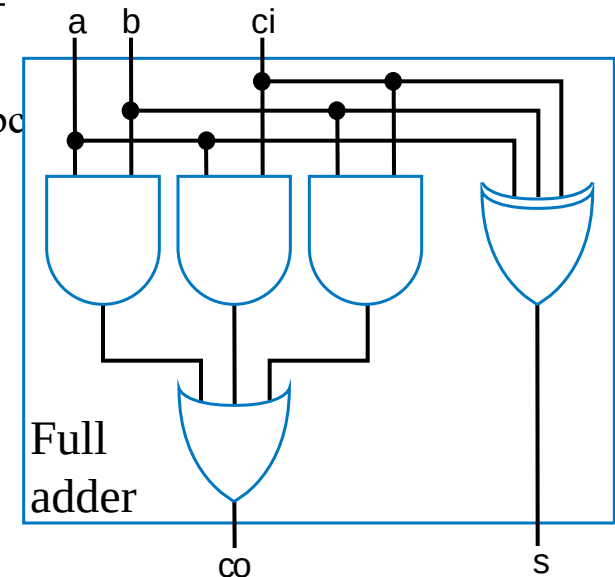
Inputs			Outputs	
a	b	ci	co	s
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Step 2: Convert to equations

$$\begin{aligned} co &= a'bc + ab'c + abc' + abc \\ co &= a'bc + abc + ab'c + abc + abc' + abc \\ co &= (a'+a)bc + (b'+b)ac + (c'+c)ab \\ co &= bc + ac + ab \end{aligned}$$

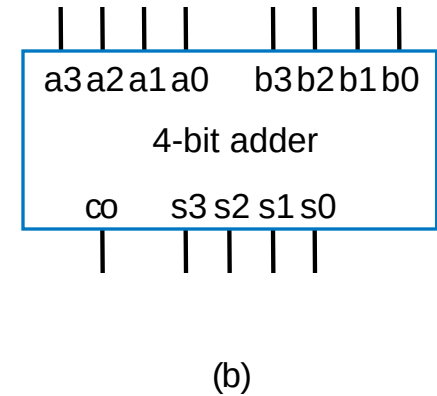
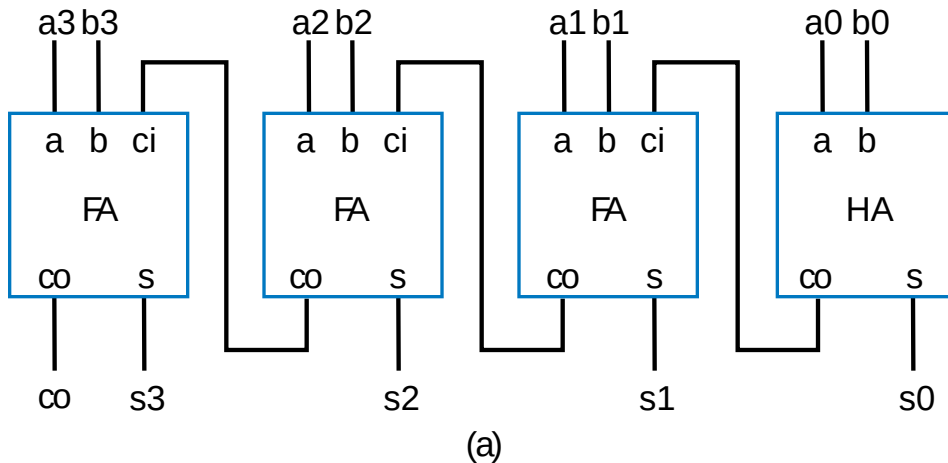
$$\begin{aligned} s &= a'b'c + a'bc' + ab'c' + abc \\ s &= a'(b'c + bc') + a(b'c' + bc) \\ s &= a'(b \text{ xor } c)' + a(b \text{ xor } c) \\ s &= a \text{ xor } b \text{ xor } c \end{aligned}$$

Step 3: Create the circuit



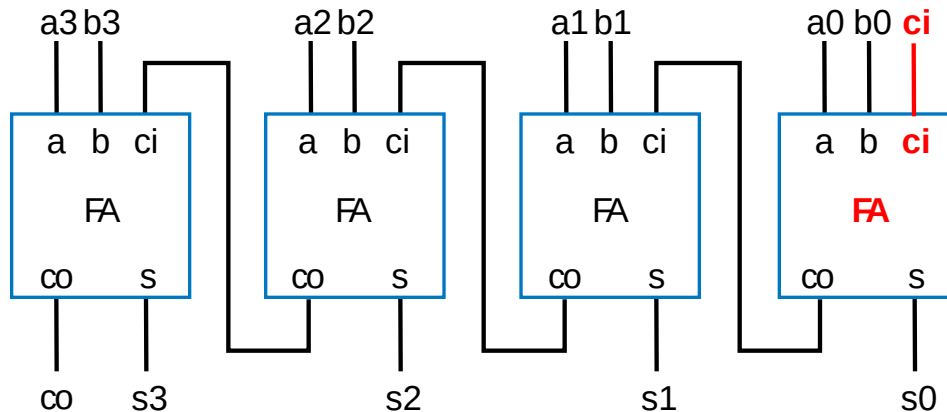
Carry-Ripple Adder

- Using half-adder and full-adders, we can build adder that adds like we would by hand
- Called a **carry-ripple adder**
 - 4-bit adder shown: Adds two 4-bit numbers, generates 5-bit output
 - 5-bit output can be considered 4-bit “sum” plus 1-bit “carry out”
 - Can easily build any size adder

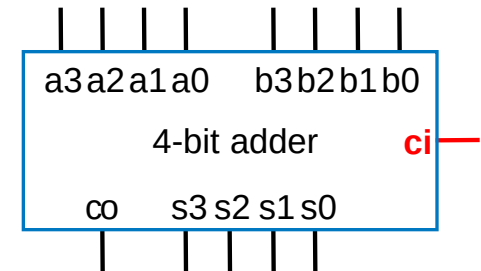


Carry-Ripple Adder

- Using full-adder instead of half-adder for first bit, we can include a “carry in” bit in the addition
 - Will be useful later when we connect smaller adders to form bigger adders



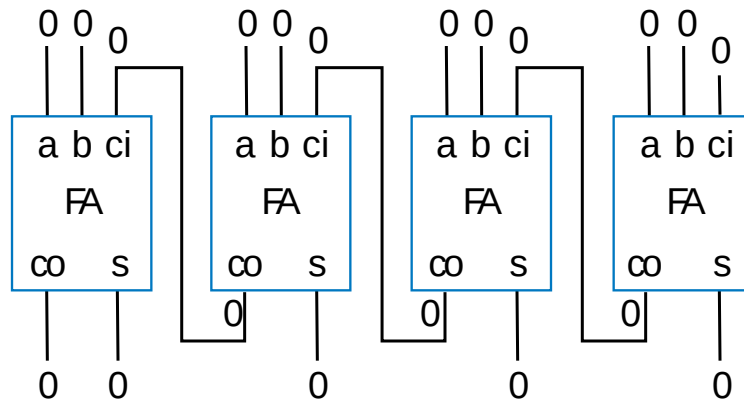
(a)



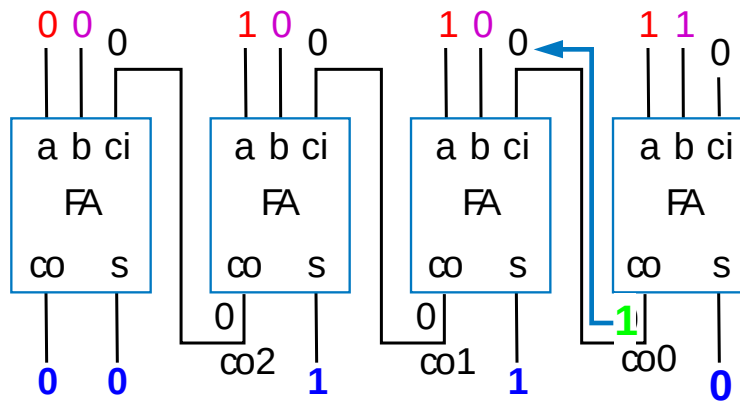
(b)



Carry-Ripple Adder's Behavior



Assume all inputs initially 0



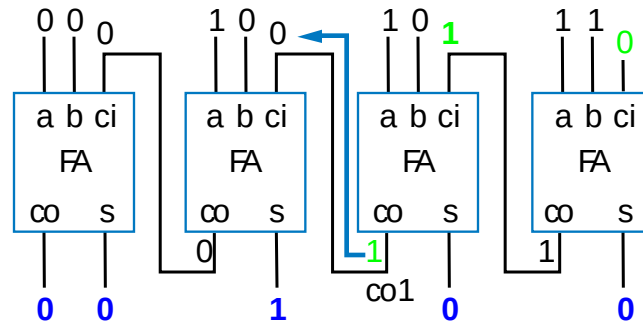
0111+0001
(answer should be 01000)

Output after 2 ns (1FA delay)

Wrong answer -- something wrong? No -- just need more time for carry to ripple through the chain of full adders.

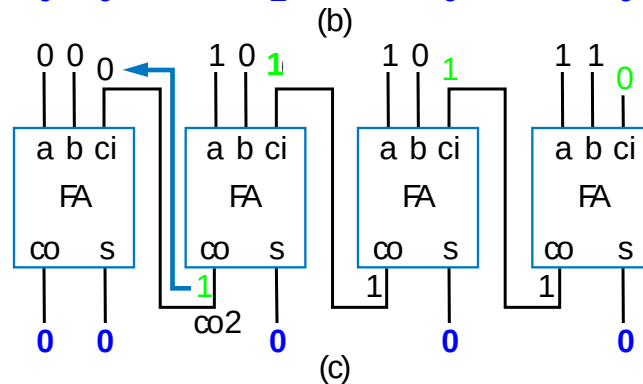


Carry-Ripple Adder's Behavior

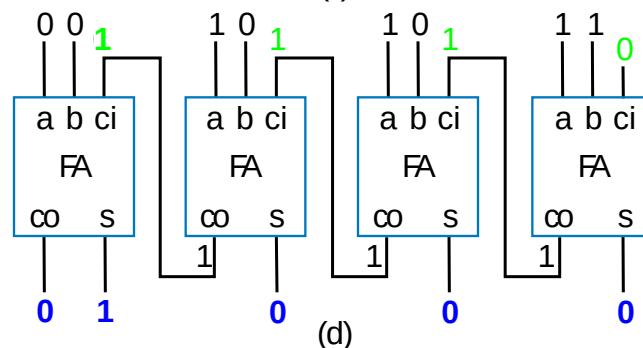


0111+0001
(answer should be 01000)

Outputs after 4ns (2 FA delays)



Outputs after 6ns (3 FA delays)

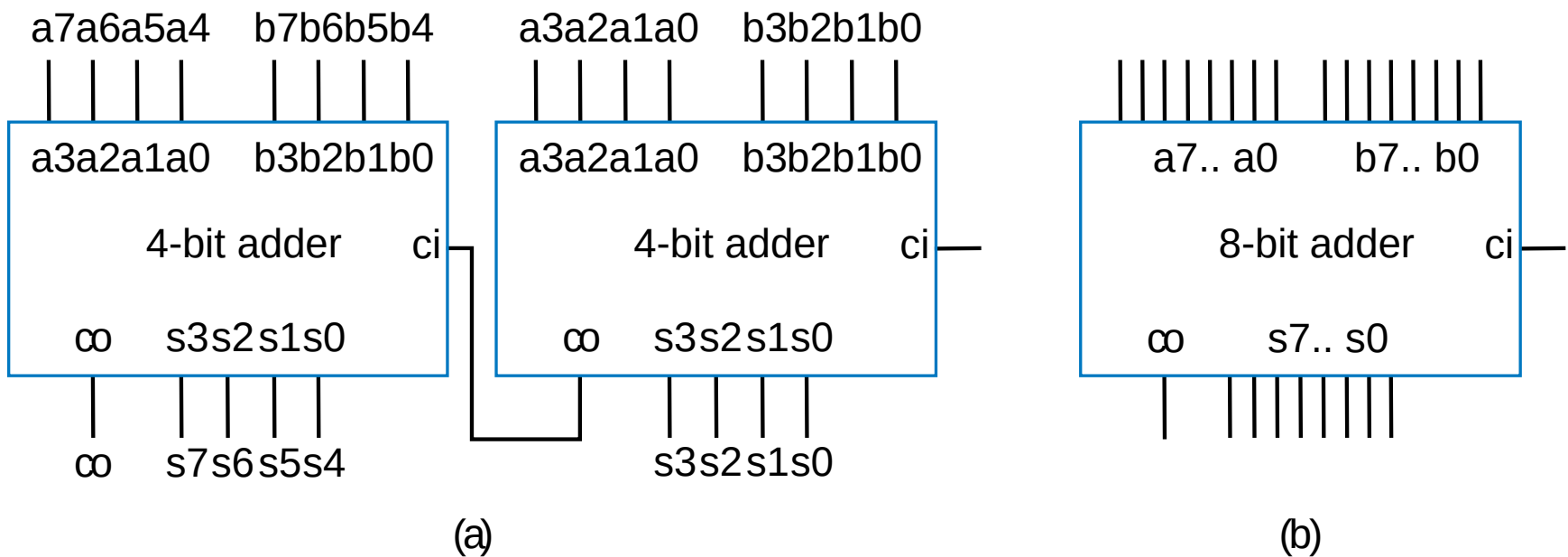


Output after 8ns (4 FA delays)

Correct answer appears after 4 FA delays

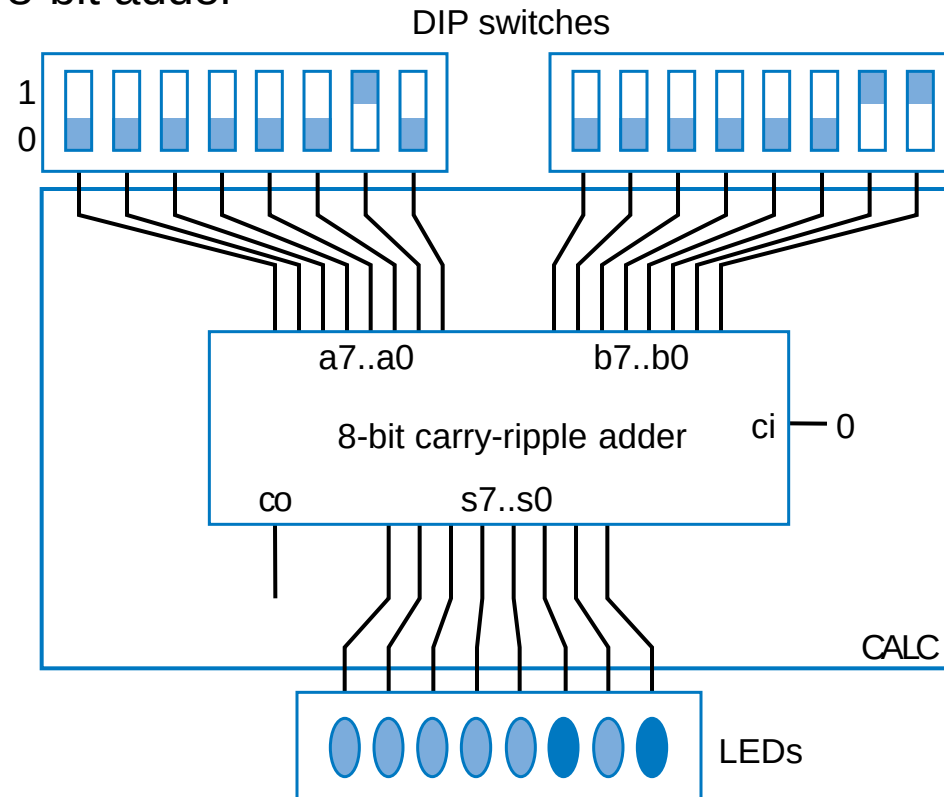


Cascading Adders



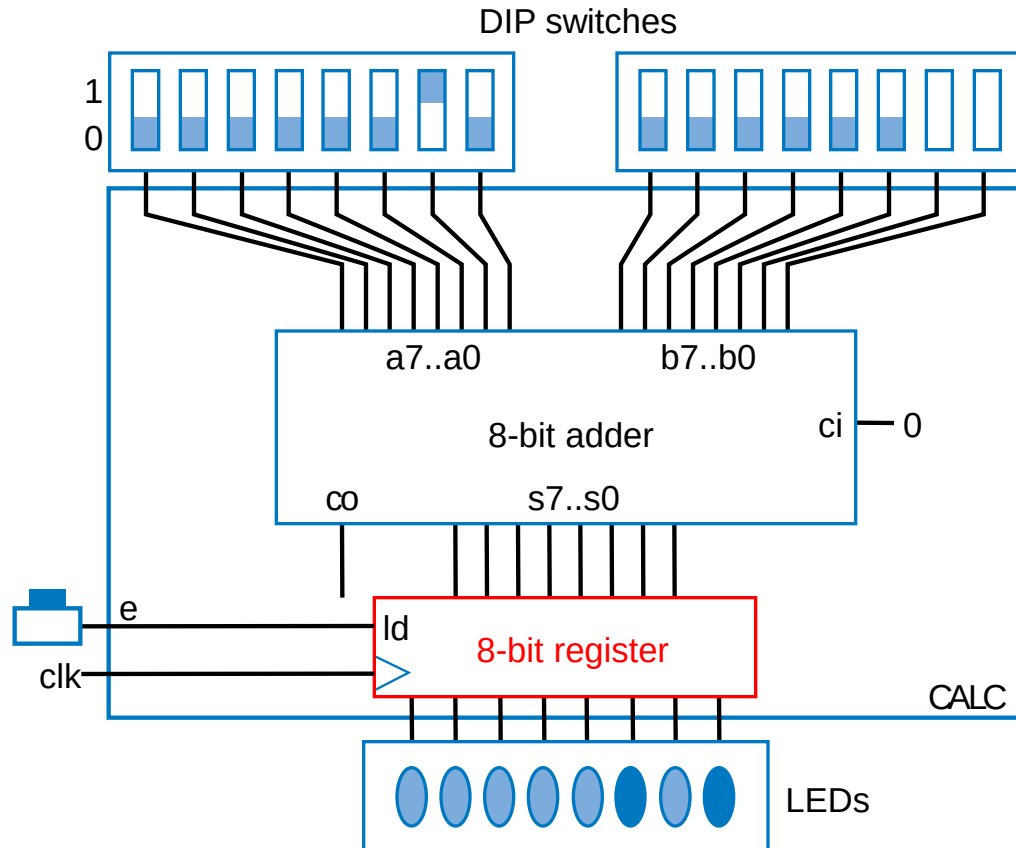
Adder Example: DIP-Switch-Based Adding Calculator

- Goal: Create calculator that adds two 8-bit binary numbers, specified using DIP switches
 - DIP switch: Dual-inline package switch, move each switch up or down
 - Solution: Use 8-bit adder



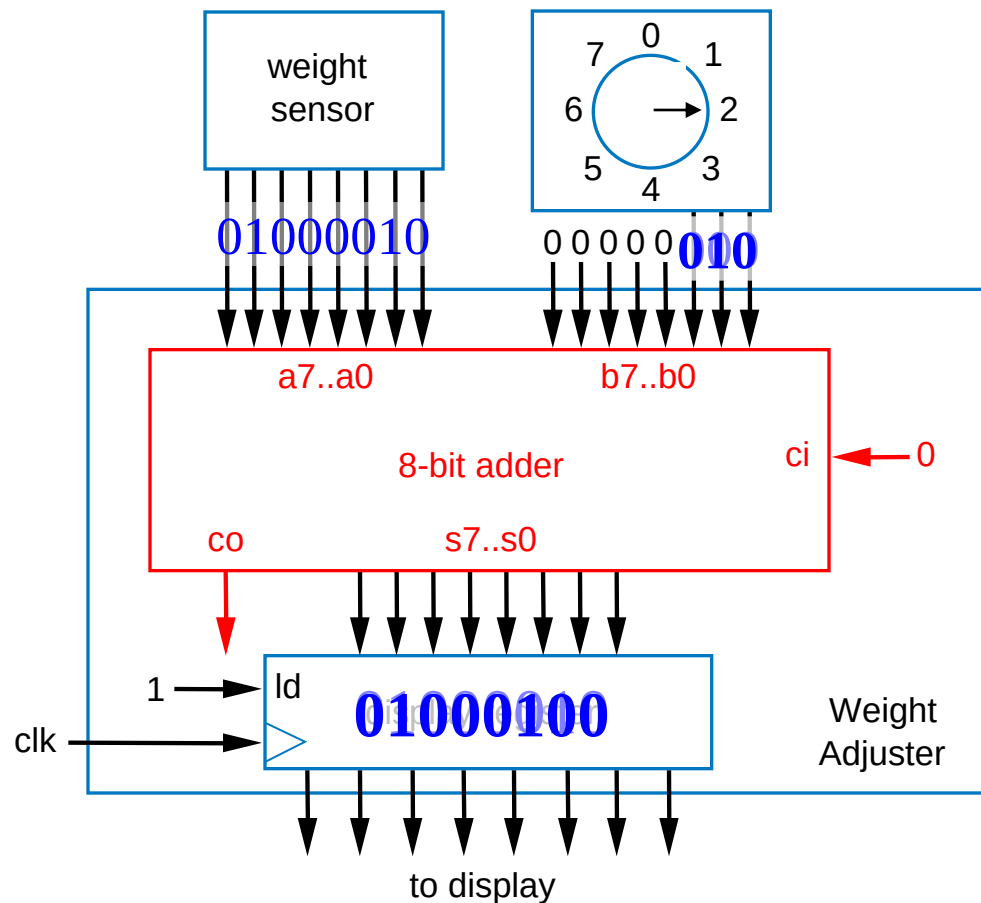
Adder Example: DIP-Switch-Based Adding Calculator

- To prevent spurious values from appearing at output, can place register at output
 - Actually, the light flickers from spurious values would be too fast for humans to detect -- but the principle of registering outputs to avoid spurious values being read by external devices (which normally aren't humans) applies here.



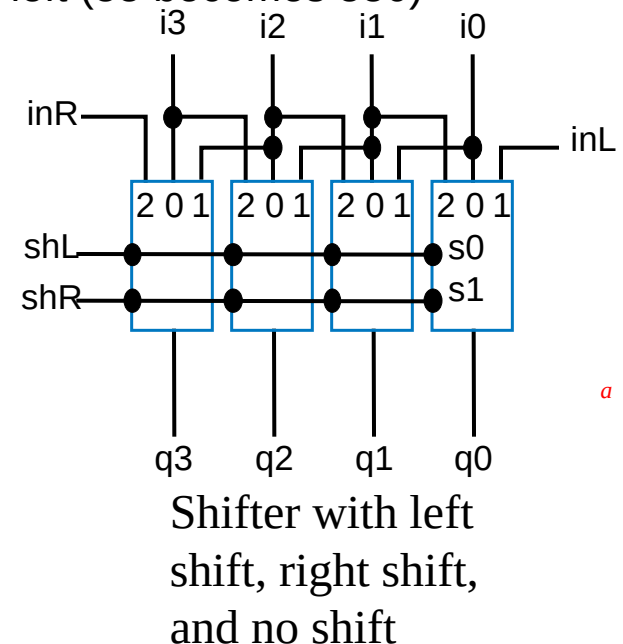
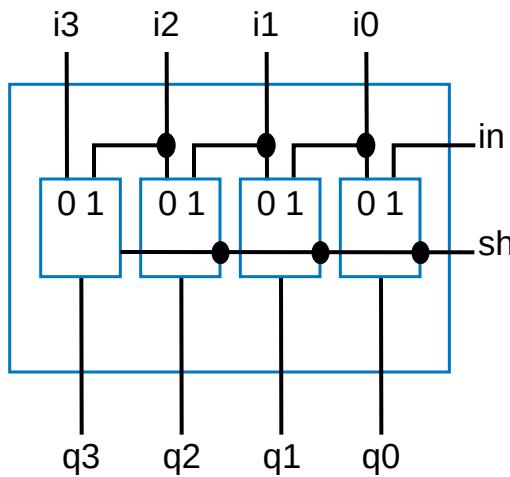
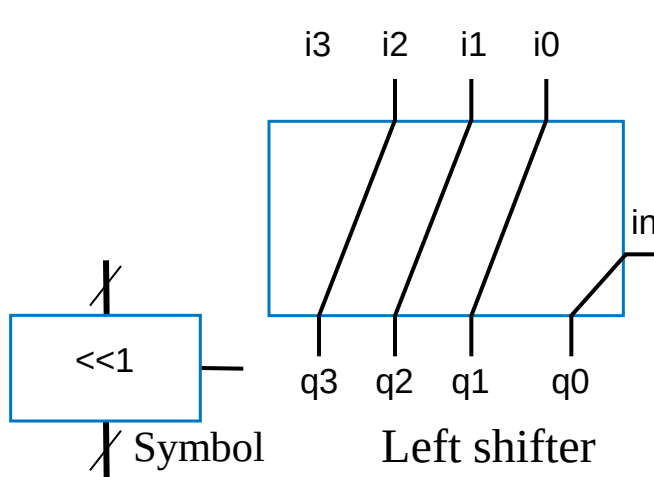
Adder Example: Compensating Weight Scale

- Weight scale with compensation amount of 0-7
 - To compensate for inaccurate sensor due to physical wear
 - Use 8-bit adder



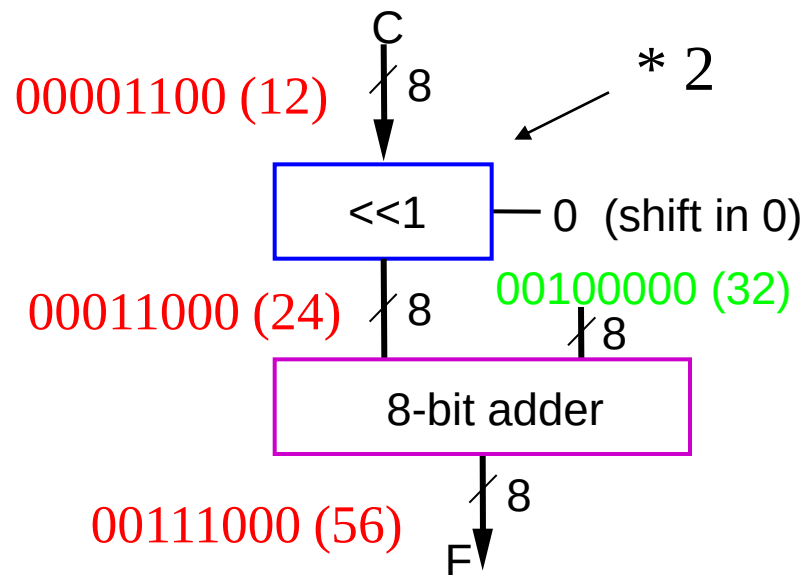
Shifters

- Shifting (e.g., left shifting 0011 yields 0110) useful for:
 - Manipulating bits
 - Converting serial data to parallel (remember earlier above-mirror display example with shift registers)
 - Shift left once is same as multiplying by 2 (0011 (3) becomes 0110 (6))
 - Why? Essentially appending a 0 -- Note that multiplying decimal number by 10 accomplished just by appending 0, i.e., by shifting left (55 becomes 550)
 - Shift right once same as dividing by 2

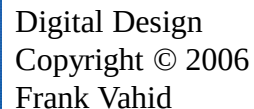


Shifter Example: Approximate Celsius to Fahrenheit Converter

- Convert 8-bit Celsius input to 8-bit Fahrenheit output
 - $F = C * 9/5 + 32$
 - Approximate: $F = C * 2 + 32$
 - Use left shift: $F = \text{left_shift}(C) + 32$

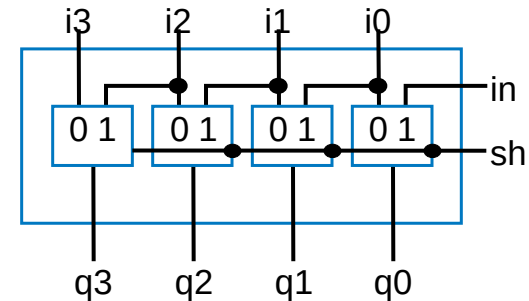


- Add, then divide by four
 - Same as shift right by 2
 - Use three adders, and right shift by two



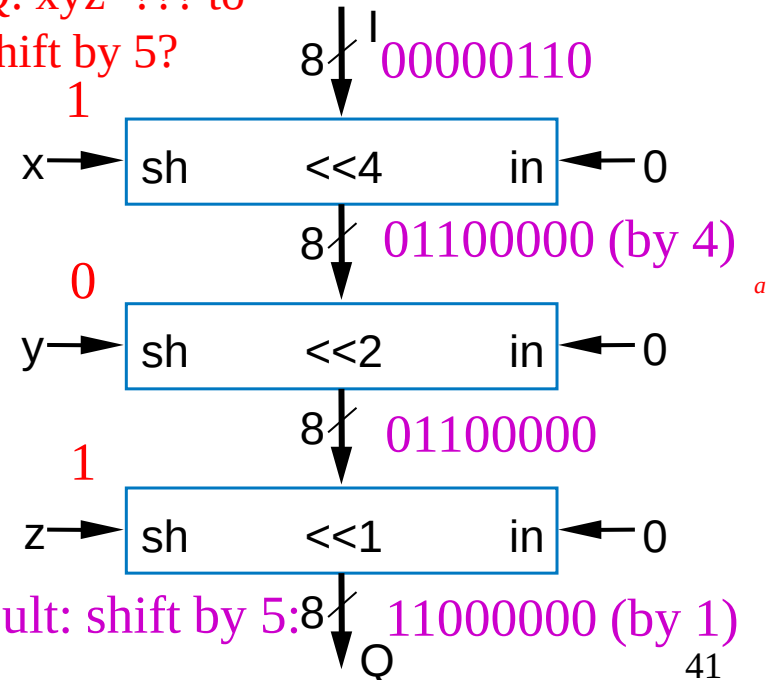
Barrel Shifter

- A shifter that can shift by any amount
 - 4-bit barrel left shift can shift left by 0, 1, 2, or 3 positions
 - 8-bit barrel left shifter can shift left by 0, 1, 2, 3, 4, 5, 6, or 7 positions
 - (Shifting an 8-bit number by 8 positions is pointless -- you just lose all the bits)
- Could design using 8x1 muxes and lots of wires
 - Too many wires
- More elegant design
 - Chain three shifters: 4, 2, and 1
 - Can achieve any shift of 0..7 by enabling the correct combination of those three shifters, i.e., shifts should sum to desired amount



Shift by 1 shifter uses 2x1 muxes. 8x1 mux solution for 8-bit barrel shifter: too many wires.

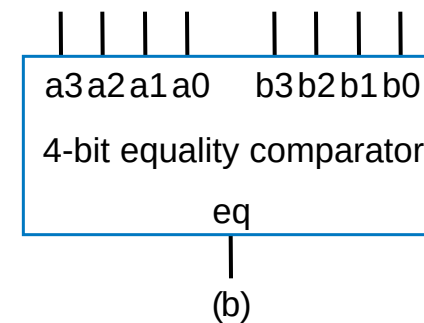
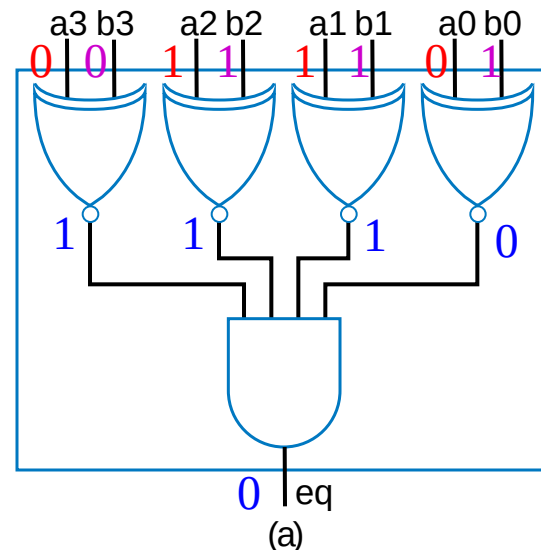
Q: xyz=??? to shift by 5?



Comparators

- ***N-bit equality comparator***: Outputs 1 if two N-bit numbers are equal
 - 4-bit equality comparator with inputs A and B
 - a_3 must equal b_3 , $a_2 = b_2$, $a_1 = b_1$, $a_0 = b_0$
 - Two bits are equal if both 1, or both 0
 - $eq = (a_3b_3 + a_3'b_3') * (a_2b_2 + a_2'b_2') * (a_1b_1 + a_1'b_1') * (a_0b_0 + a_0'b_0')$
 - Recall that XNOR outputs 1 if its two input bits are the same
 - $eq = (a_3 \text{ xnor } b_3) * (a_2 \text{ xnor } b_2) * (a_1 \text{ xnor } b_1) * (a_0 \text{ xnor } b_0)$

0110 = 0111 ?



Magnitude Comparator

- ***N-bit magnitude comparator:***

Indicates whether $A > B$, $A = B$, or $A < B$, for its two N-bit inputs A and B

- How design? Consider how compare by hand. First compare a_3 and b_3 . If equal, compare a_2 and b_2 . And so on. Stop if comparison not equal -- whichever's bit is 1 is greater. If never see unequal bit pair, $A = B$.

A=1011 B=1001

1011 **1**001 **Equal**

1**0**11 1**0**01 **Equal**

10**1**1 10**0**1 **Unequal**

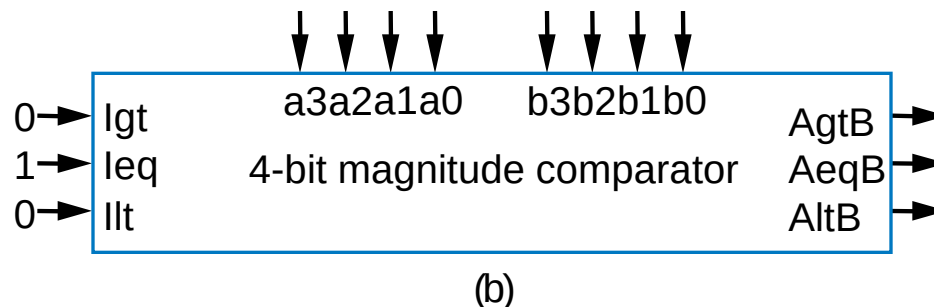
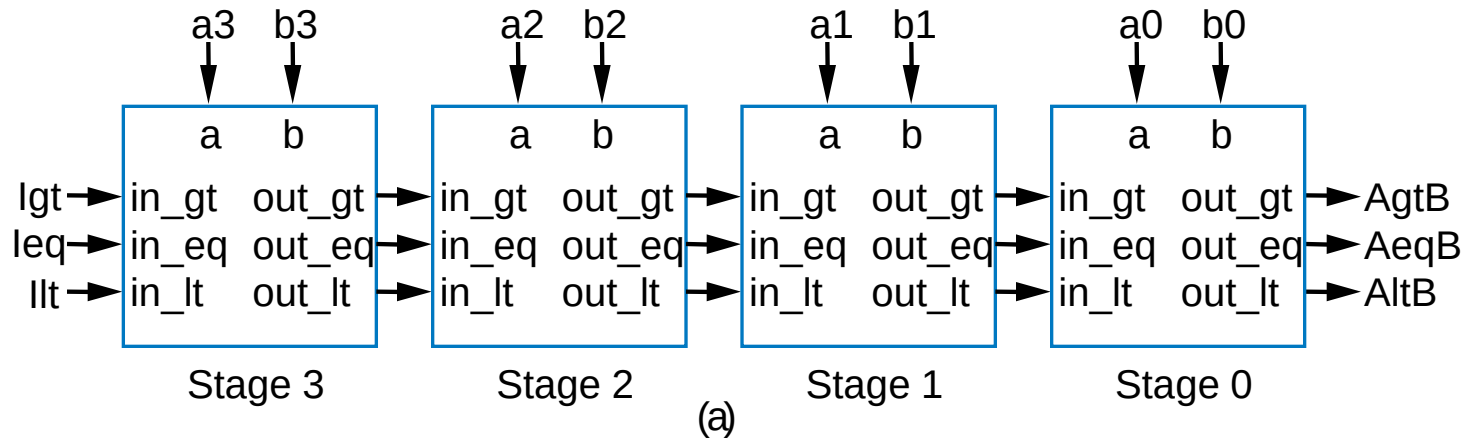
So $A > B$

a

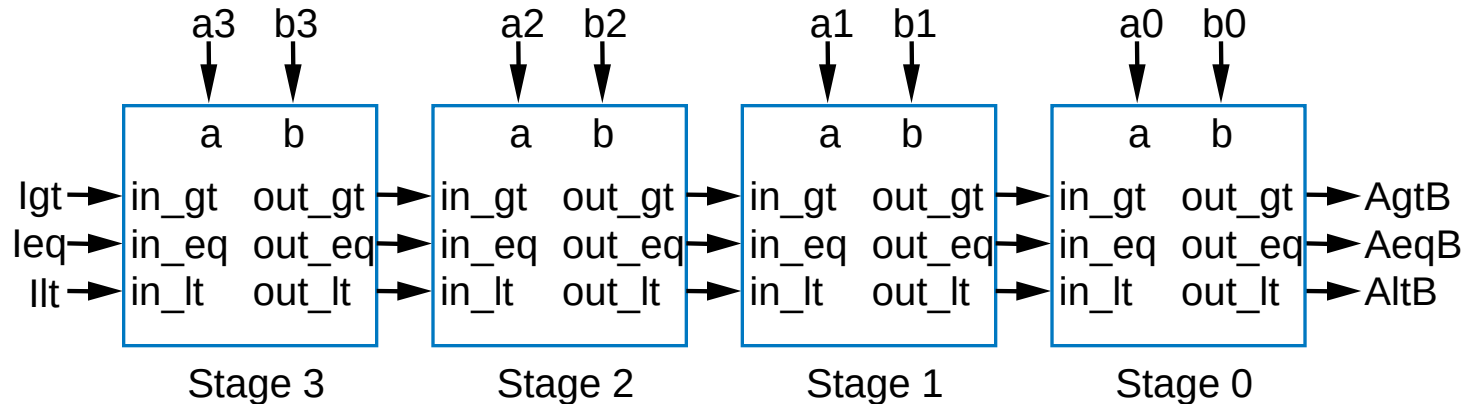


Magnitude Comparator

- By-hand example leads to idea for design
 - Start at left, compare each bit pair, pass results to the right
 - Each bit pair called a stage
 - Each stage has 3 inputs indicating results of higher stage, passes results to lower stage



Magnitude Comparator

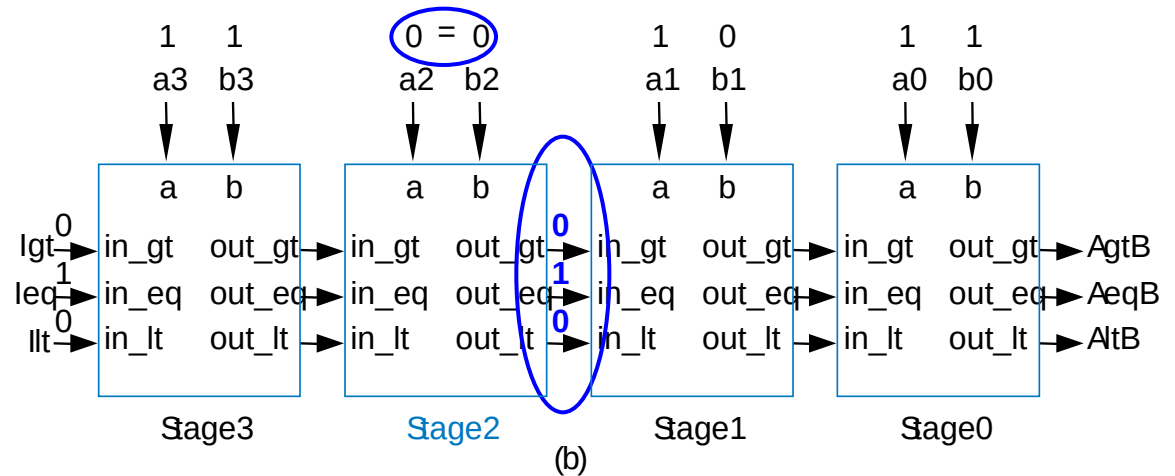
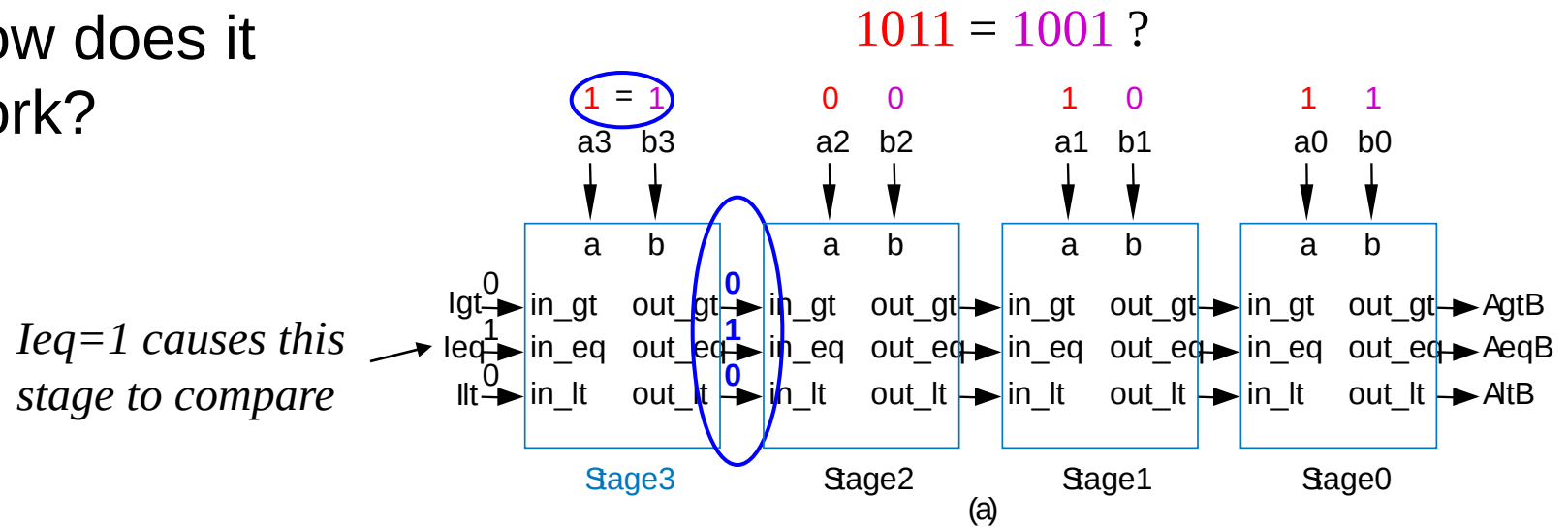


- Each stage:
 - $out_gt = in_gt + (in_eq * a * b')$
 - $A > B$ (so far) if already determined in higher stage, or if higher stages equal but in this stage $a=1$ and $b=0$
 - $out_lt = in_lt + (in_eq * a' * b)$
 - $A < B$ (so far) if already determined in higher stage, or if higher stages equal but in this stage $a=0$ and $b=1$
 - $out_eq = in_eq * (a \text{ XNOR } b)$
 - $A = B$ (so far) if already determined in higher stage and in this stage $a=b$ too
 - Simple circuit inside each stage, just a few gates (not shown)



Magnitude Comparator

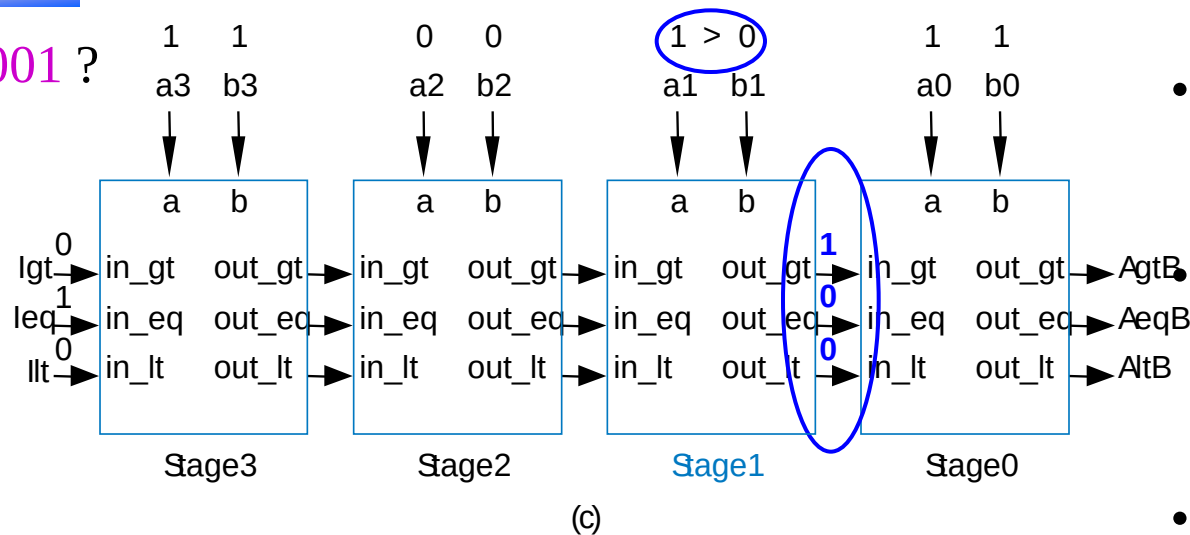
- How does it work?



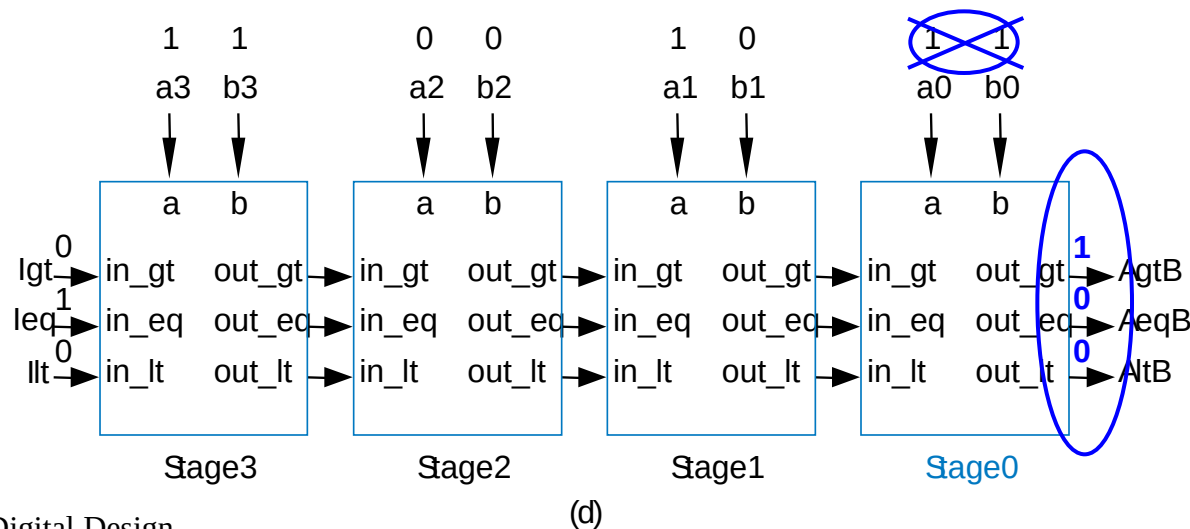
Magnitude Comparator



1011 = 1001 ?



a



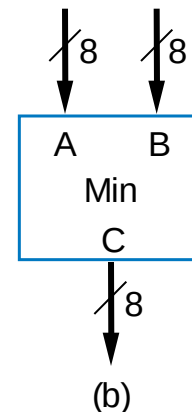
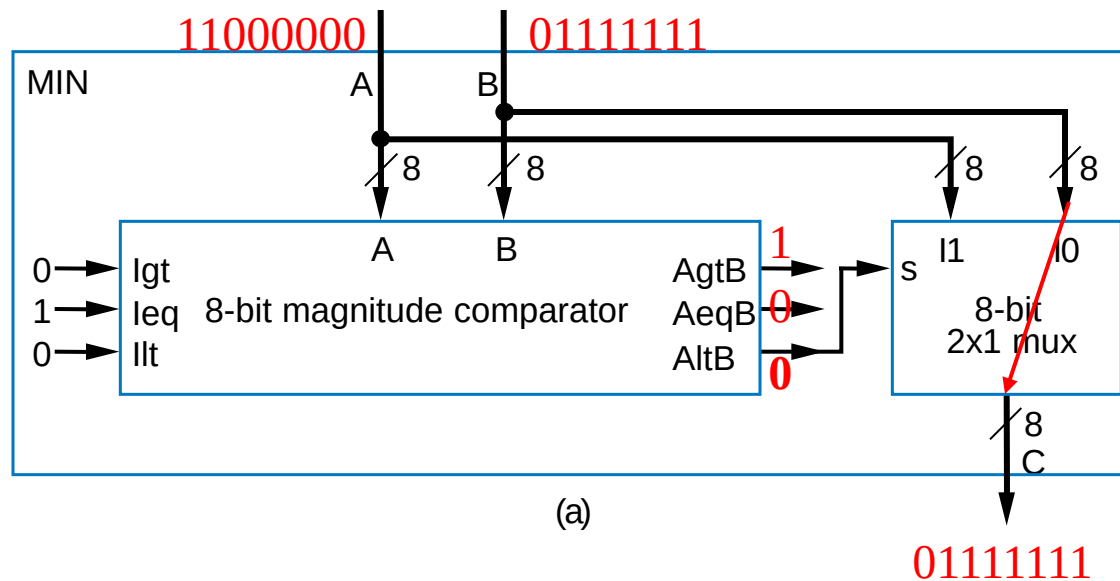
- Final answer appears on the right
- Takes time for answer to “ripple” from left to right
- Thus called “carry-ripple style” after the carry-ripple adder
 - Even though there’s no “carry” involved



Magnitude Comparator Example:

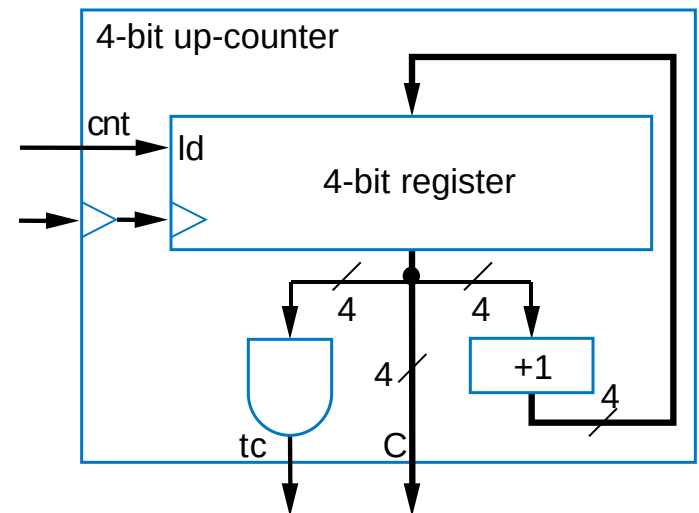
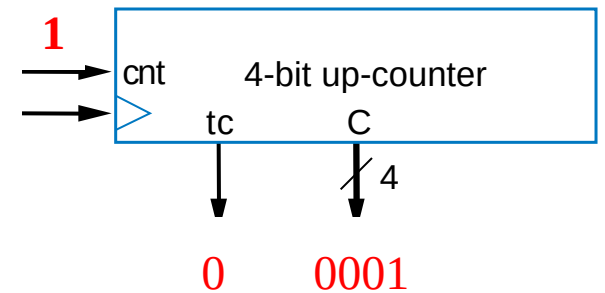
Minimum of Two Numbers

- Design a combinational component that computes the minimum of two 8-bit numbers
 - Solution: Use 8-bit magnitude comparator and 8-bit 2x1 mux
 - If $A < B$, pass A through mux. Else, pass B.



Counters

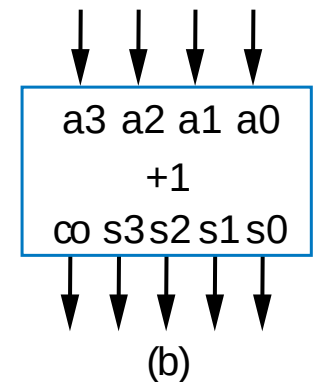
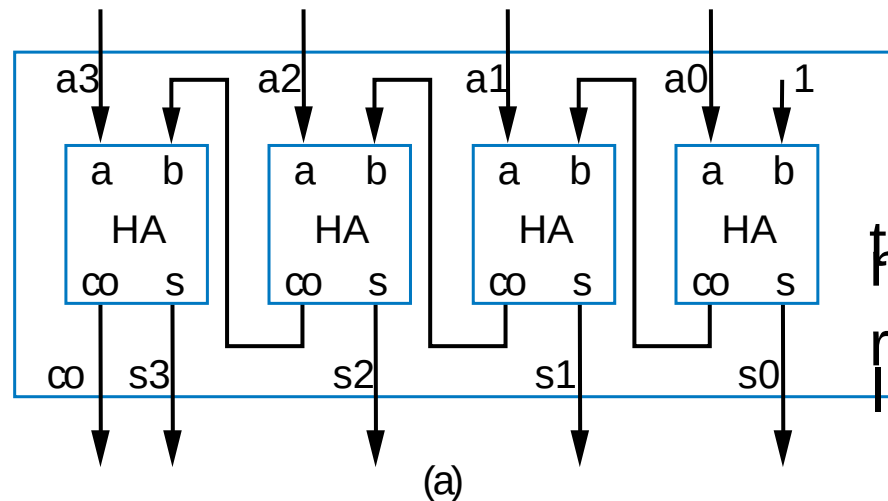
- ***N-bit up-counter***: N-bit register that can increment (add 1) to its own value on each clock cycle
 - 0000, 0001, 0010, 0011, ..., 1110, 1111, 0000
 - Note how count “rolls over” from 1111 to 0000
 - Terminal (last) count, tc, equals 1 during value just before rollover
- Internal design
 - Register, incrementer, and N-input AND gate to detect terminal count



Incrementer

- Counter design used incrementer
- Incrementer design
 - Could use carry-ripple adder with B input set to 00...001
 - But when adding 00...001 to another number, the leading 0's obviously don't need to be considered -- so just two bits being added per column
 - Use half-adders (adds two bits) rather than full-adders (adds three bits)

carries: 0 1 1
 0 0 1 1
 unused + 1
 00100



Incrementer

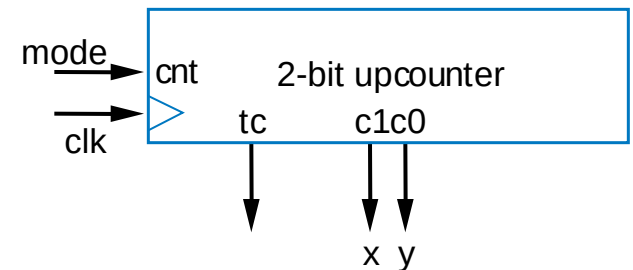
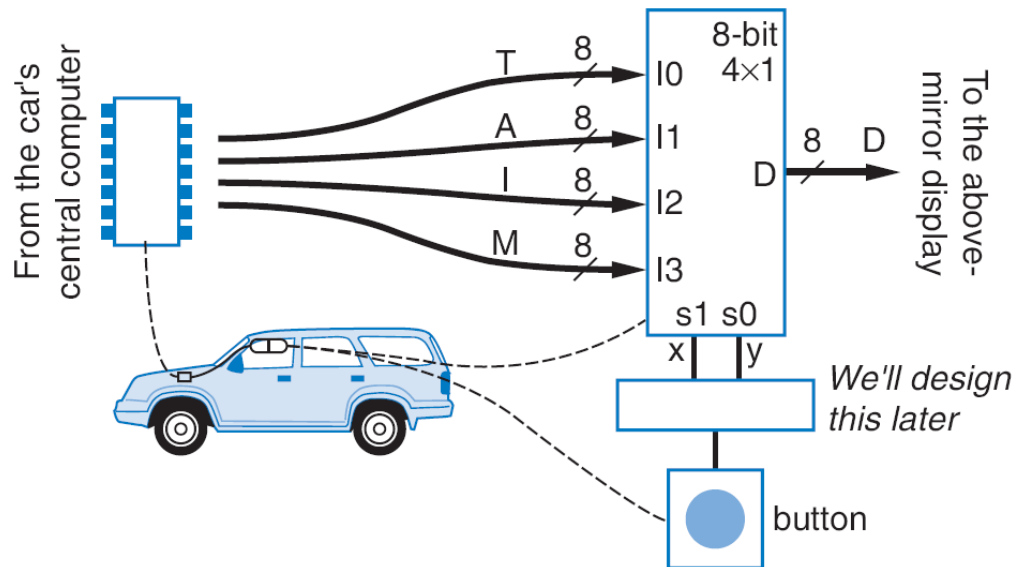
- Can build faster incrementer using combinational logic design process
 - Capture truth table
 - Derive equation for each output
 - $c0 = a3a2a1a0$
 - ...
 - $s0 = a0'$
 - Results in small and fast circuit
 - Note: works for small N -- larger N leads to exponential growth, like for N-bit adder

Inputs				Outputs				
a3	a2	a1	a0	c0	s3	s2	s1	s0
0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	1	0
0	0	1	0	0	0	0	1	1
0	0	1	1	0	0	1	0	0
0	1	0	0	0	0	1	0	1
0	1	0	1	0	0	1	1	0
0	1	1	0	0	0	1	1	1
0	1	1	1	0	1	0	0	0
1	0	0	0	0	1	0	0	1
1	0	0	1	0	1	0	1	0
1	0	1	0	0	1	0	1	1
1	0	1	1	0	1	1	0	0
1	1	0	0	0	1	1	0	1
1	1	0	1	0	1	1	1	0
1	1	1	0	0	1	1	1	1
1	1	1	1	1	0	0	0	0



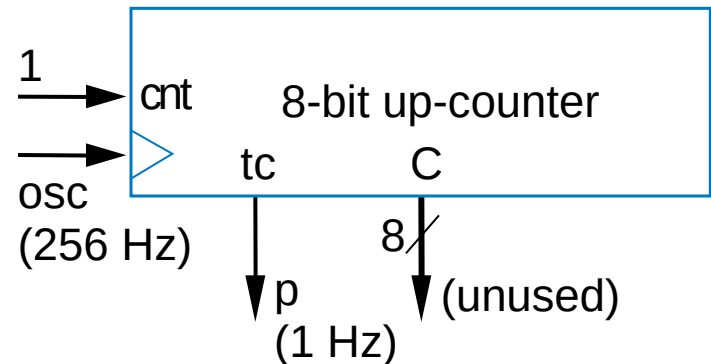
Counter Example: Mode in Above-Mirror Display

- Recall above-mirror display example from Chapter 2
 - Assumed component that incremented xy input each time button pressed: 00, 01, 10, 11, 00, 01, 10, 11, 00, ...
 - Can use 2-bit up-counter
 - Assumes mode=1 for just one clock cycle during each button press
 - Recall “Button press synchronizer” example from Chapter 3



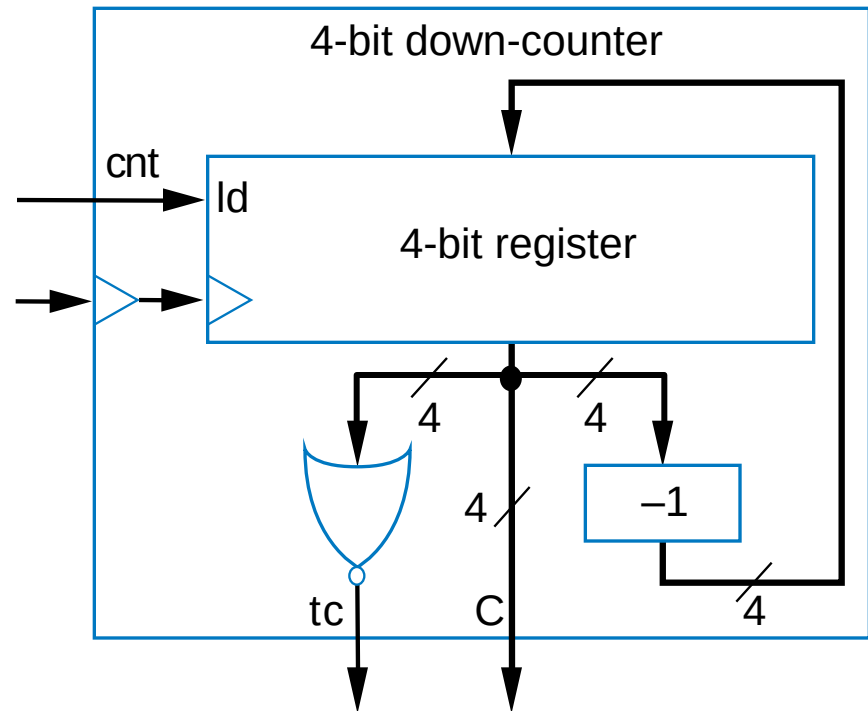
Counter Example: 1 Hz Pulse Generator Using 256 Hz Oscillator

- Suppose have 256 Hz oscillator, but want 1 Hz pulse
 - 1 Hz is 1 pulse per second -- useful for keeping time
 - Design using 8-bit up-counter, use tc output as pulse
 - Counts from 0 to 255 (256 counts), so pulses tc every 256 cycles



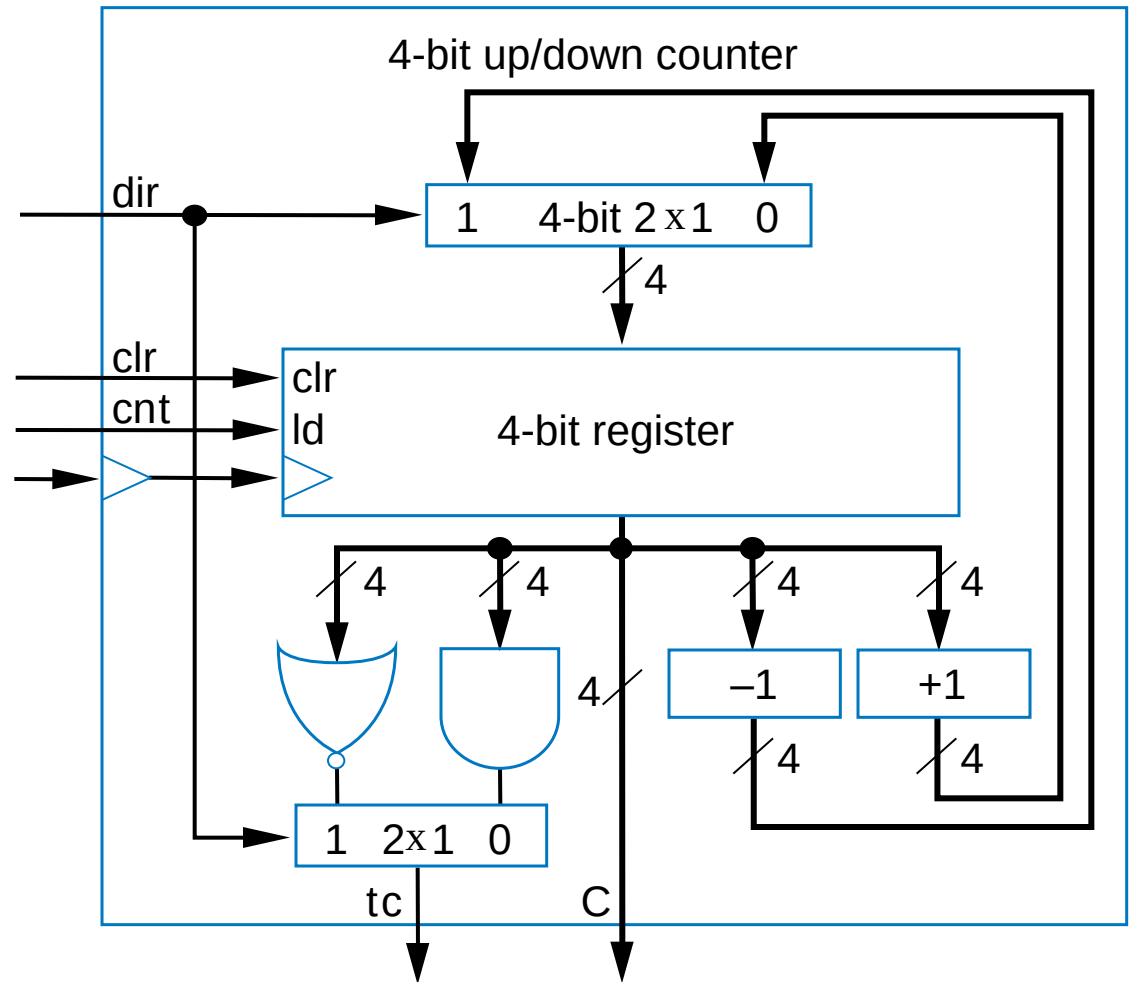
Down-Counter

- 4-bit down-counter
 - 1111, 1110, 1101, 1100, ..., 0011, 0010, 0001, 0000, 1111, ...
 - Terminal count is 0000
 - Use NOR gate to detect
 - Need decrementer (-1) – design like designed incrementer



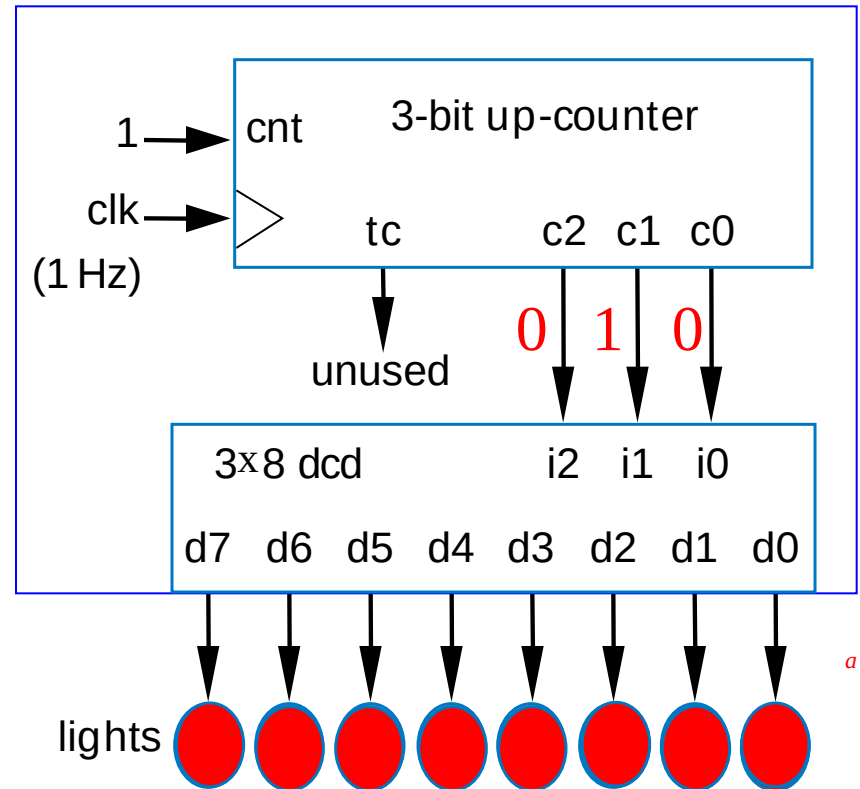
Up/Down-Counter

- Can count either up or down
 - Includes both incrementer and decrements
 - Use dir input to select, using 2x1: dir=0 means up
 - Likewise, dir selects appropriate terminal count value

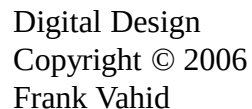
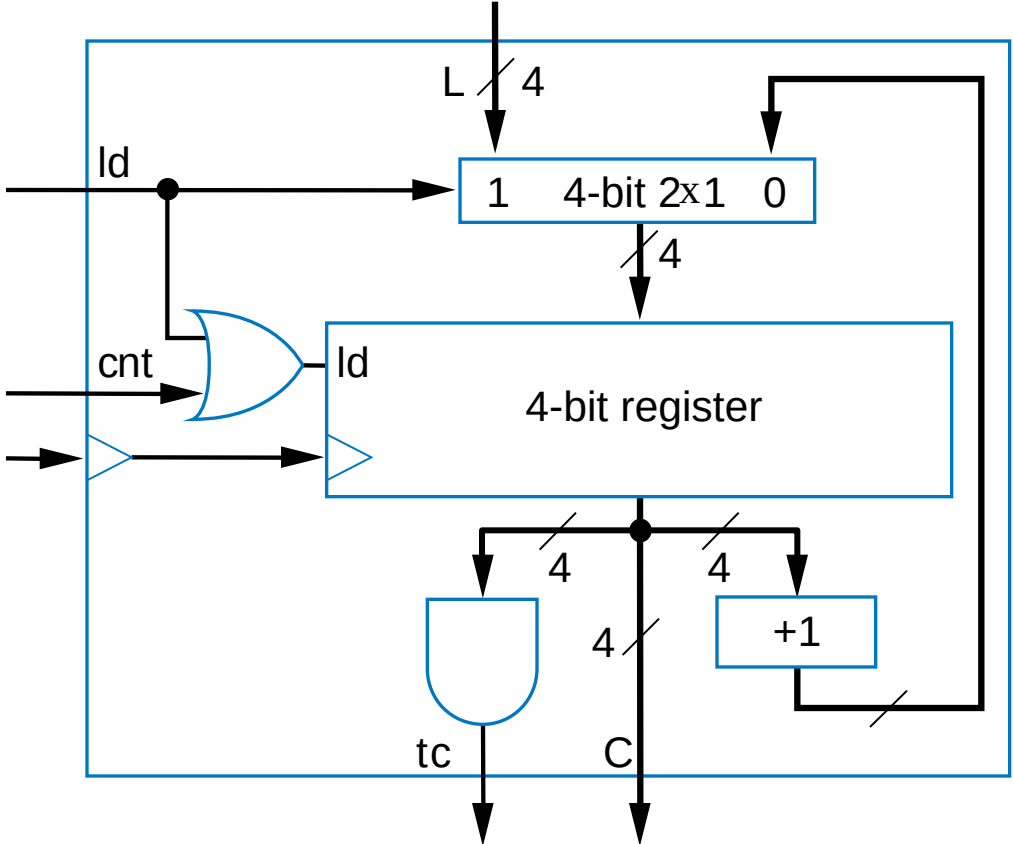


Counter Example: Light Sequencer

- Illuminate 8 lights from right to left, one at a time, one per second
- Use 3-bit up-counter to counter from 0 to 7
- Use 3x8 decoder to illuminate appropriate light
- Note: Used **3-bit** counter with 3x8 decoder
 - *NOT* an 8-bit counter – why not?

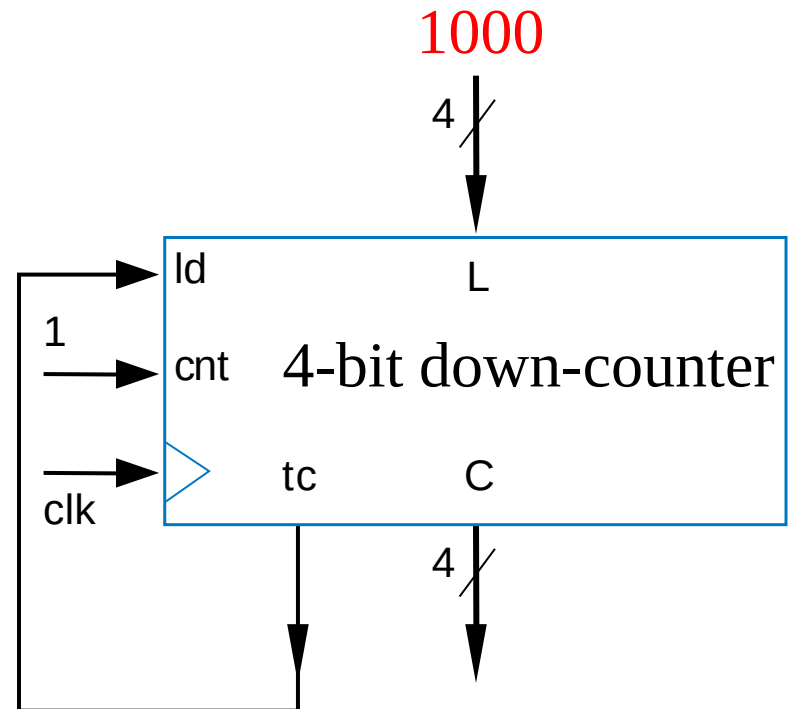


- Designed using 2x1 mux
 - Id input selects incremented value or external value
- Load the internal register when loading external value or when counting



Counter with Parallel Load

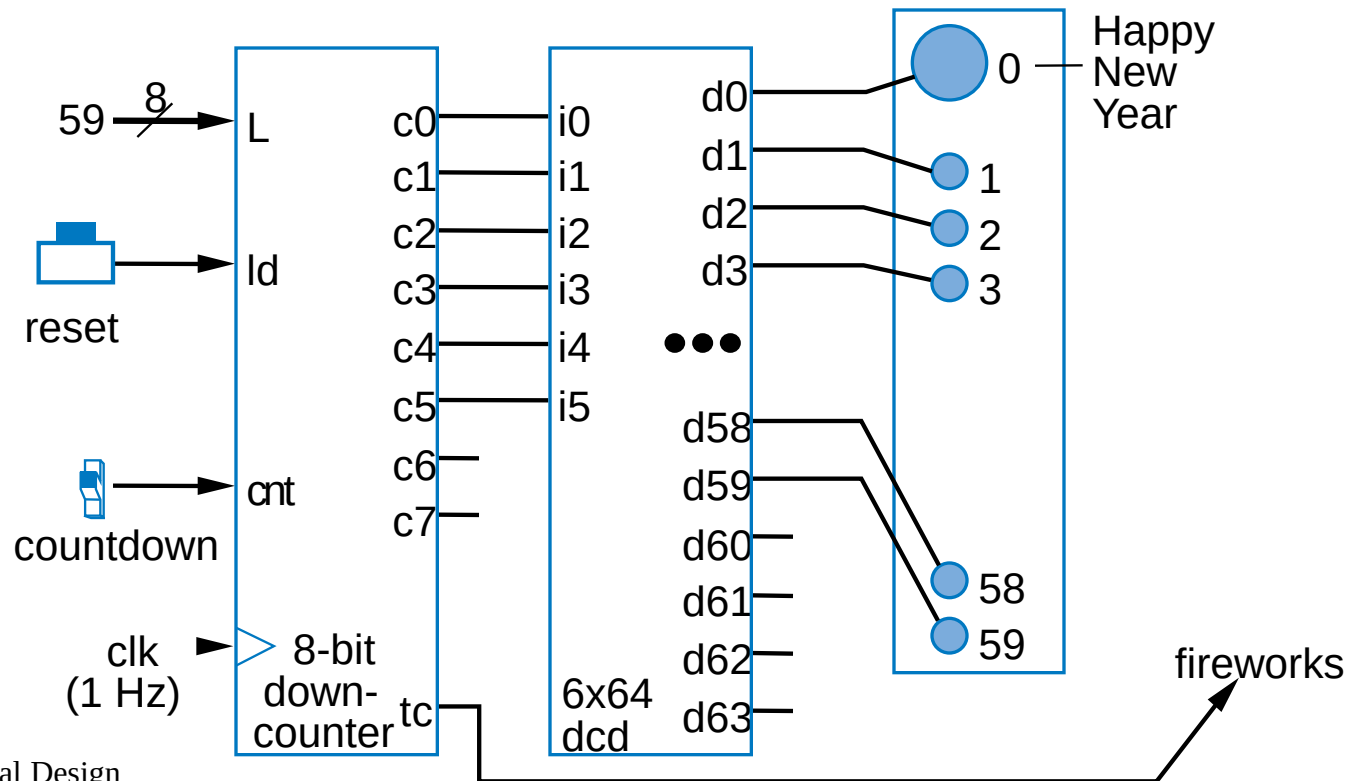
- Useful to create pulses at specific multiples of clock
 - Not just at N-bit counter's natural wrap-around of 2^N
- Example: Pulse every 9 clock cycles
 - Use 4-bit down-counter with parallel load
 - Set parallel load input to 8 (1000)
 - Use terminal count to reload
 - When count reaches 0, next cycle loads 8.
 - Why load 8 and not 9? Because 0 is included in count sequence:
 - 8, 7, 6, 5, 4, 3, 2, 1, 0 → 9 counts



Counter Example:

New Year's Eve Countdown Display

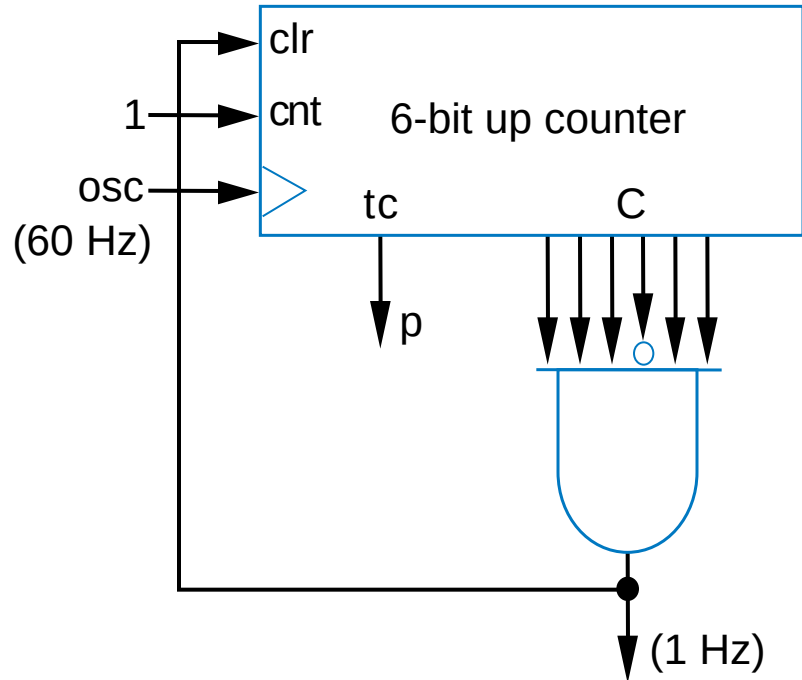
- Chapter 2 example previously used microprocessor to counter from 59 down to 0 in binary
- Can use 8-bit (or 7- or 6-bit) down-counter instead, initially loaded with 59



Counter Example:

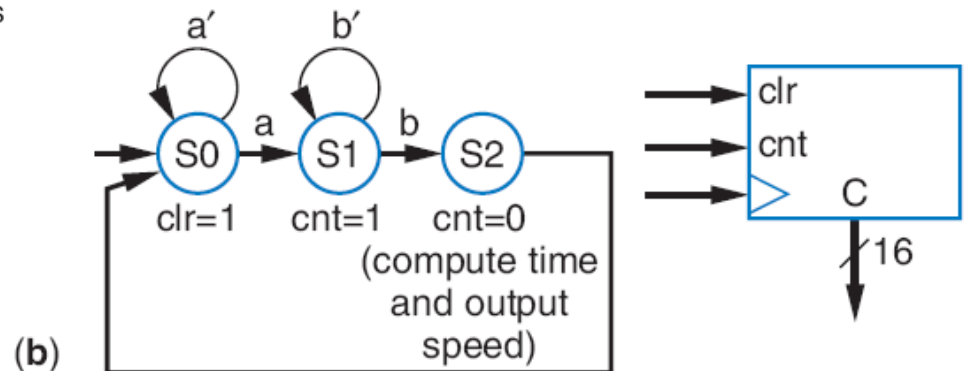
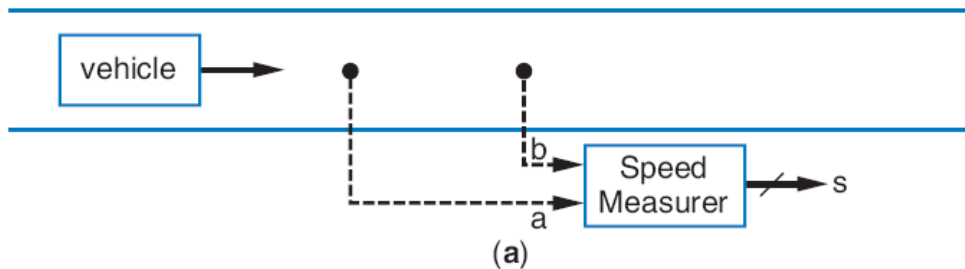
1 Hz Pulse Generator from 60 Hz Clock

- U.S. electricity standard uses 60 Hz signal
 - Device may convert that to 1 Hz signal to count seconds
- Use 6-bit up-counter
 - Can count from 0 to 63
 - Create simple logic to detect 59 (for 60 counts)
 - Use to clear the counter back to 0 (or to load 0)



Timer

- A type of counter used to measure time
 - If we know the counter's clock frequency and the count, we know the time that's been counted
- Example: Compute car's speed using two sensors
 - First sensor (a) clears and starts timer
 - Second sensor (b) stops timer
 - Assuming clock of 1kHz, timer output represents time to travel between sensors. Knowing the distance, we can compute speed



Multiplier – Array Style

- Can build multiplier that mimics multiplication by hand
 - Notice that multiplying multiplicand by 1 is same as ANDing with 1

0110	(the top number is called the <i>multiplicand</i>)
0011	(the bottom number is called the <i>multiplier</i>)
----	(each row below is called a <i>partial product</i>)
0110	(because the rightmost bit of the multiplier is 1, and $0110 * 1 = 0110$)
0110	(because the second bit of the multiplier is 1, and $0110 * 1 = 0110$)
0000	(because the third bit of the multiplier is 0, and $0110 * 0 = 0000$)
+0000	(because the leftmost bit of the multiplier is 0, and $0110 * 0 = 0000$)

00010010	(the <i>product</i> is the sum of all the partial products: 18, which is $6 * 3$)



Multiplier – Array Style

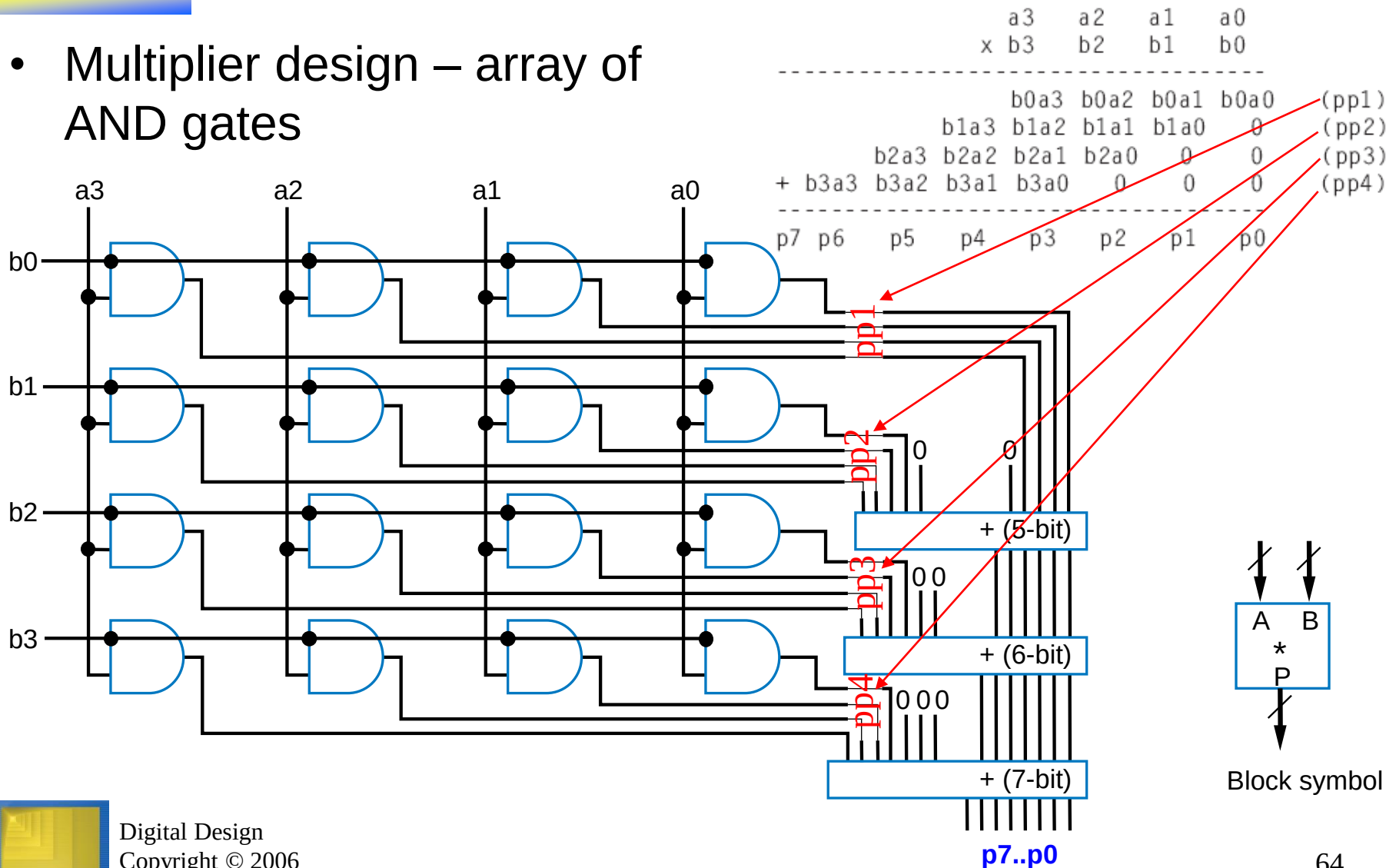
- Generalized representation of multiplication by hand

$$\begin{array}{r}
 \begin{array}{cccc}
 & a3 & a2 & a1 & a0 \\
 x & b3 & b2 & b1 & b0
 \end{array} \\
 \hline
 & & & b0a3 & b0a2 & b0a1 & b0a0 & (pp1) \\
 & & b1a3 & b1a2 & b1a1 & b1a0 & 0 & (pp2) \\
 & b2a3 & b2a2 & b2a1 & b2a0 & 0 & 0 & (pp3) \\
 + & b3a3 & b3a2 & b3a1 & b3a0 & 0 & 0 & 0 & (pp4) \\
 \hline
 p7 & p6 & p5 & p4 & p3 & p2 & p1 & p0
 \end{array}$$



Multiplier – Array Style

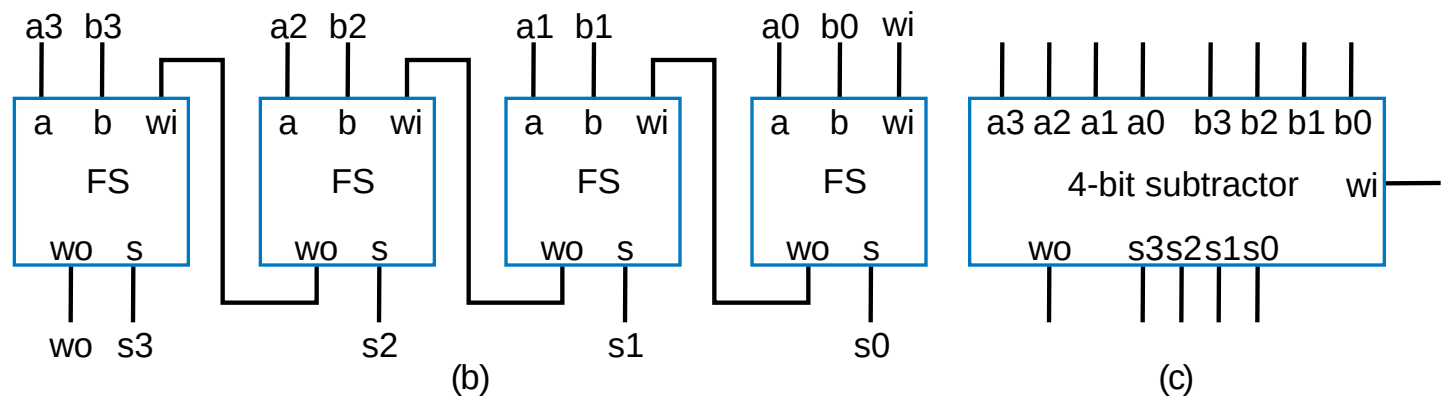
- Multiplier design – array of AND gates



Subtractor

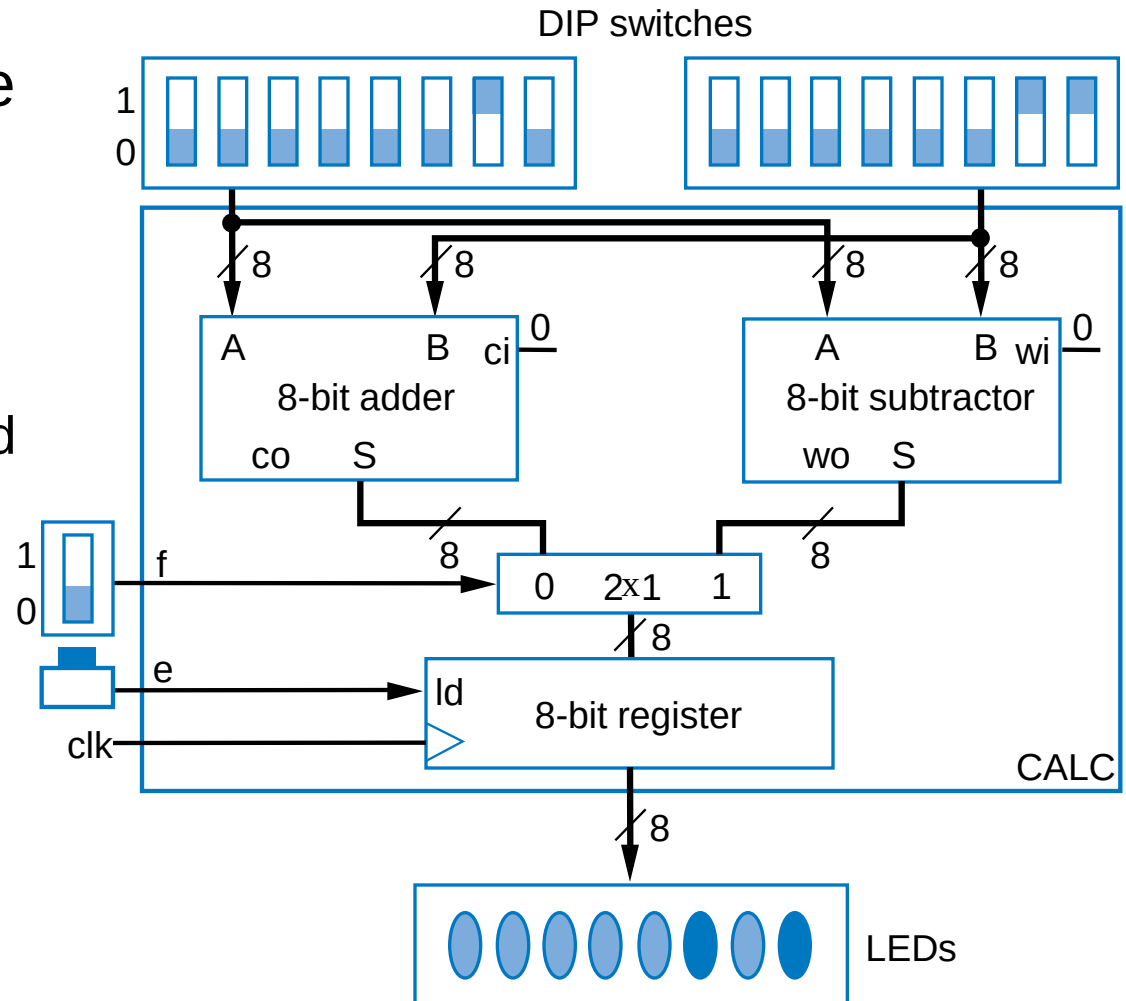
- Can build subtractor as we built carry-ripple adder
 - Mimic subtraction by hand
 - Compute borrows from columns on left
 - Use full-subtractor component:
 - wi is borrow by column on right, wo borrow from column on left

1st column	2nd column	3rd column	4th column
$\begin{array}{r} 1011 \\ - 0111 \\ \hline 1 \end{array}$	$\begin{array}{r} 1011 \\ - 0111 \\ \hline 11 \end{array}$	$\begin{array}{r} 1011 \\ - 0111 \\ \hline 011 \end{array}$	$\begin{array}{r} 1011 \\ - 0111 \\ \hline 0011 \end{array}$



Subtractor Example: DIP-Switch Based Adding/Subtracting Calculator

- Extend earlier calculator example
 - Switch f indicates whether want to add ($f=0$) or subtract ($f=1$)
 - Use subtractor and 2x1 mux



Subtractor Example:

Color Space Converter – RGB to CMYK

- Color

- Often represented as weights of three colors: red, green, and blue (RGB)
 - Perhaps 8 bits each, so specific color is 24 bits
 - White: R=11111111, G=11111111, B=11111111
 - Black: R=00000000, G=00000000, B=00000000
 - Other colors: values in between, e.g., R=00111111, G=00000000, B=00001111 would be a reddish purple
- Good for computer monitors, which mix red, green, and blue lights to form all colors



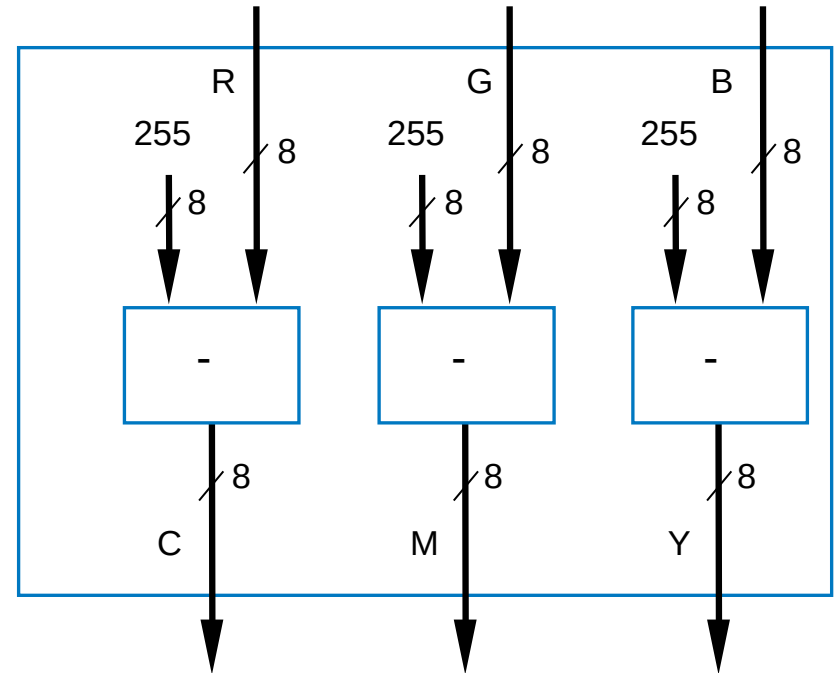
- Printers use opposite color scheme
 - Because inks *absorb* light
 - Use complementary colors of RGB: Cyan (absorbs red), reflects green and blue, Magenta (absorbs green), and Yellow (absorbs blue)



Subtractor Example:

Color Space Converter – RGB to CMYK

- Printers must quickly convert RGB to CMY
 - $C=255-R$, $M=255-G$, $Y=255-B$
 - Use subtractors as shown



Subtractor Example:

Color Space Converter – RGB to CMYK

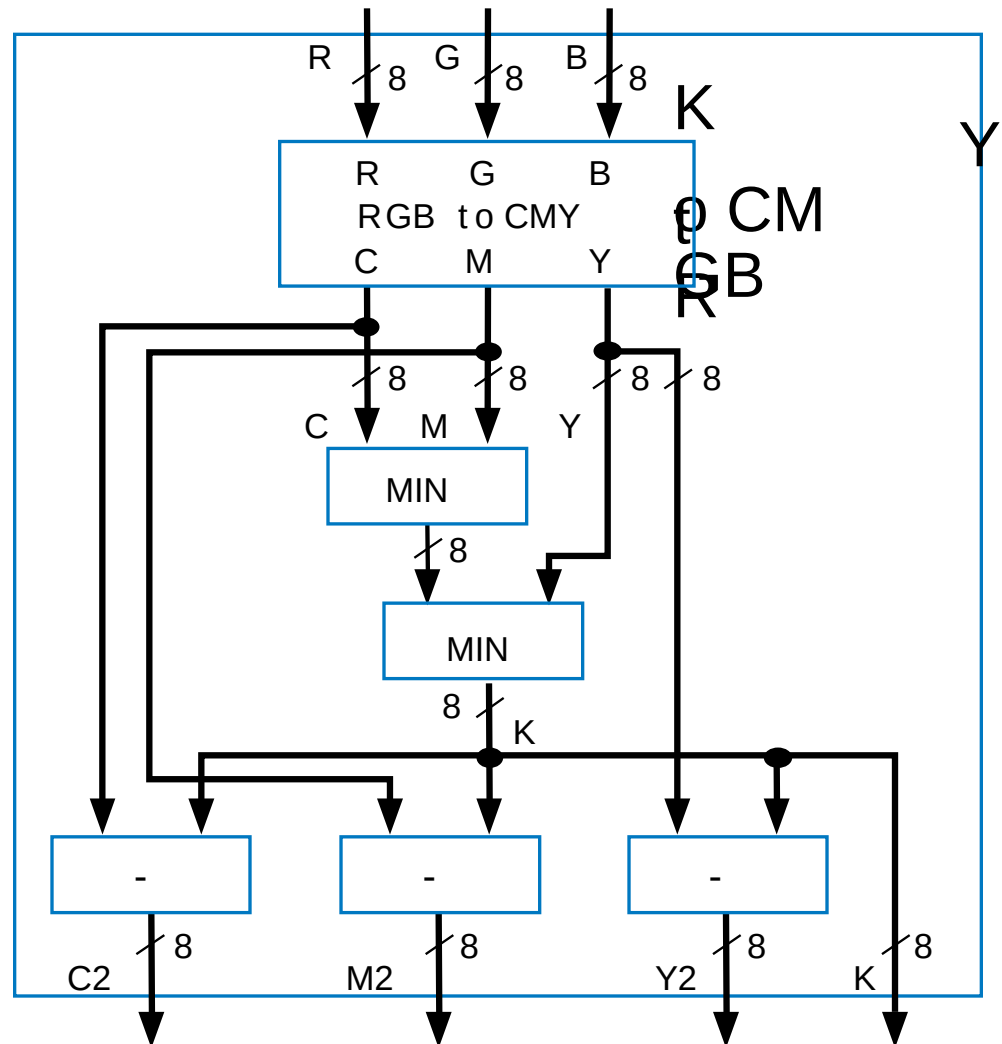
- Try to save colored inks
 - Expensive
 - Imperfect – mixing C, M, Y doesn't yield good-looking black
- Solution: Factor out the black or gray from the color, print that part using black ink
 - e.g., CMY of $(250, 200, 200) = (200, 200, 200) + (50, 0, 0)$.
 - $(200, 200, 200)$ is a dark gray – use black ink



Subtractor Example:

Color Space Converter – RGB to CMYK

- Call black part K
 - (200,200,200): K=200
 - (Letter “B” already used for blue)
- Compute minimum of C, M, Y values
 - Use MIN component designed earlier, using comparator and mux, to compute K
 - Output resulting K value, and subtract K value from C, M, and Y values
 - Ex: Input of (250,200,200) yields output of (50,0,0,200)



Representing Negative Numbers: Two's Complement

- Negative numbers common
 - How represent in binary?
- Signed-magnitude
 - Use leftmost bit for sign bit
 - So -5 would be:
 - 1101 using four bits
 - 10000101 using eight bits
- Better way: Two's complement
 - Big advantage: Allows us to perform subtraction using addition
 - Thus, only need adder component, no need for separate subtractor component!



Ten's Complement

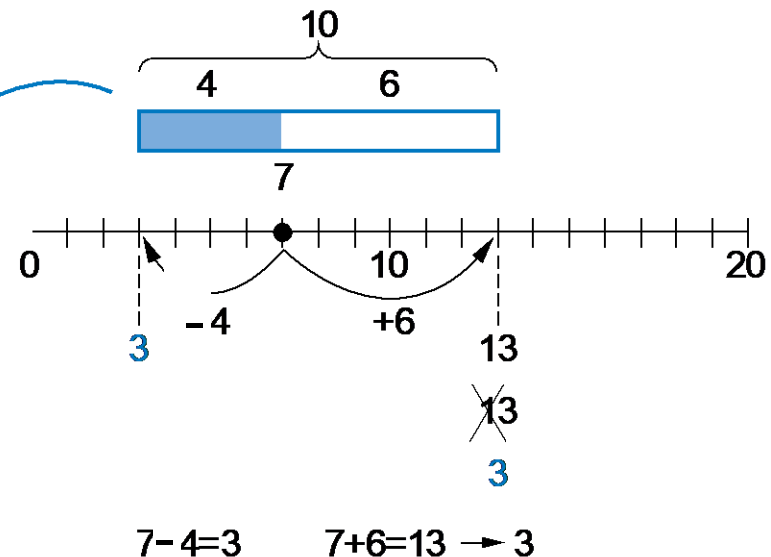
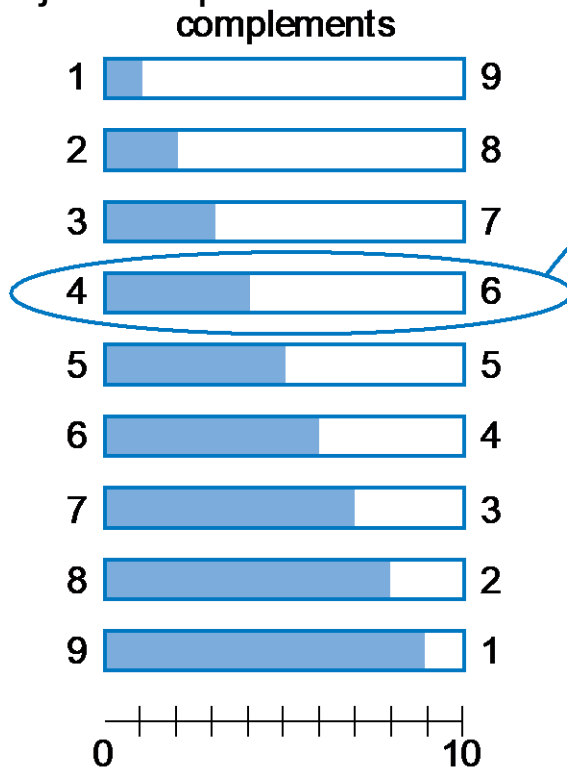
- Before introducing two's complement, let's consider ten's complement
 - But, be aware that computers DO NOT USE TEN'S COMPLEMENT. Introduced for intuition only.
 - Complements for each base ten number shown to right – Complement is the number that when added results in 10

1	→	9
2	→	8
3	→	7
4	→	6
5	→	5
6	→	4
7	→	3
8	→	2
9	→	1



Ten's Complement

- Nice feature of ten's complement
 - Instead of subtracting a number, adding its complement results in answer exactly 10 too much
 - So just drop the 1 – results in subtracting using addition only



Adding the complement results in an answer exactly 10 too much – dropping the tens column gives the right answer.



Two's Complement is Easy to Compute: Just Invert Bits and Add 1

- Hold on!
 - Sure, adding the ten's complement achieves subtraction using addition only
 - But don't we have to perform *subtraction* to have determined the complement in the first place? e.g., we only know that the complement of 4 is 6 by subtracting $10-4=6$ in the first place.
- True – but in binary, it turns out that the two's complement can be computed **easily**
 - Two's complement of 011 is 101, because $011 + 101$ is 1000
 - Could compute complement of 011 as $1000 - 011 = 101$
 - **Easier method: Just invert all the bits, and add 1**
 - The complement of 011 is $100+1 = 101$ -- it works!

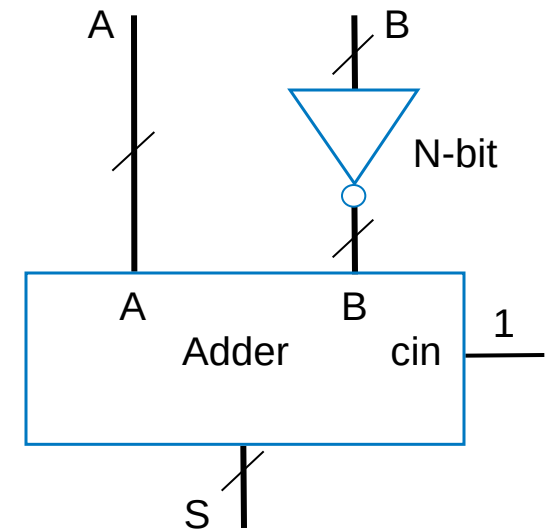
Q: What is the two's complement of 0101? A: $1010+1=1011$ a
(check: $0101+1011=10000$)

Q: What is the two's complement of 0011? A: $1100+1=1101$



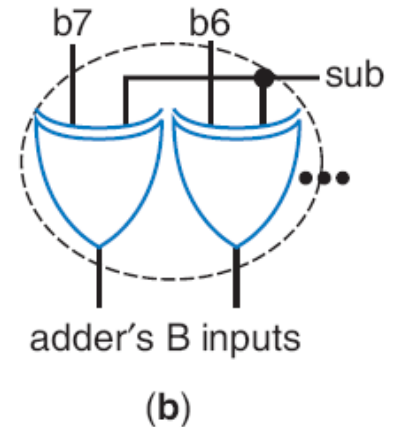
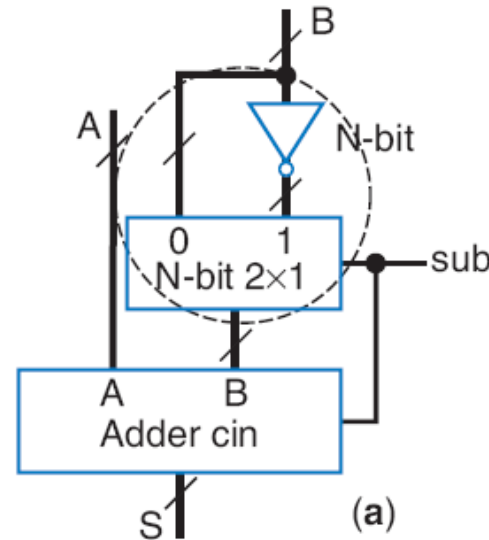
Two's Complement Subtractor Built with an Adder

- Using two's complement
$$A - B = A + (-B)$$
$$= A + (\text{two's complement of } B)$$
$$= A + \text{invert_bits}(B) + 1$$
- So build subtractor using adder by inverting B's bits, and setting carry in to 1



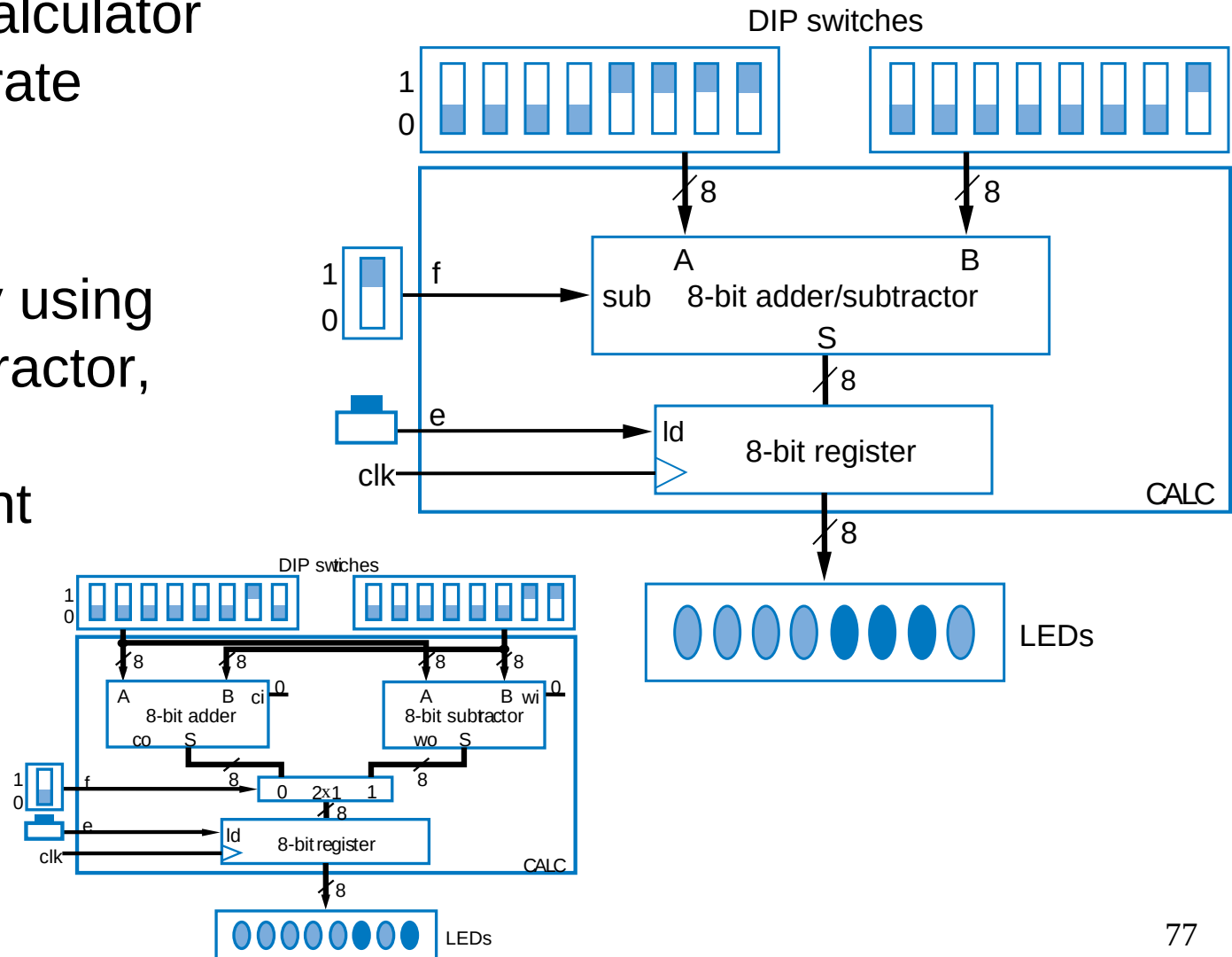
Adder/Subtractor

- Adder/subtractor: control input determines whether add or subtract
 - Can use 2x1 mux – sub input passes either B or inverted B
 - Alternatively, can use XOR gates – if sub input is 0, B's bits pass through; if sub input is 1, XORs invert B's bits



Adder/Subtractor Example: Calculator

- Previous calculator used separate adder and subtractor
- Improve by using adder/subtractor, and two's complement numbers



Overflow

- Sometimes result can't be represented with given number of bits
 - Either too large magnitude of positive or negative
 - e.g., 4-bit two's complement addition of $0111 + 0001$ ($7 + 1 = 8$). But 4-bit two's complement can't represent number > 7
 - $0111 + 0001 = 1000$ WRONG answer, 1000 in two's complement is -8, not +8
 - Adder/subtractor should indicate when overflow has occurred, so result can be discarded



Detecting Overflow: Method 1

- Assuming 4-bit two's complement numbers, can detect overflow by detecting when the two numbers' sign bits are the same but are different from the result's sign bit
 - If the two numbers' sign bits are different, overflow is impossible
 - Adding a positive and negative can't exceed largest magnitude positive or negative
- Simple circuit
 - $\text{overflow} = a_3'b_3's_3 + a_3b_3s_3'$
 - Include "overflow" output bit on adder/subtractor

sign bits

$\begin{array}{r} \textcircled{0} \ 1 \ 1 \ 1 \\ + \textcircled{0} \ 0 \ 0 \ 1 \\ \hline \textcircled{1} \ 0 \ 0 \ 0 \end{array}$	$\begin{array}{r} \textcircled{1} \ 1 \ 1 \ 1 \\ + \textcircled{1} \ 0 \ 0 \ 0 \\ \hline \textcircled{0} \ 1 \ 1 \ 1 \end{array}$	$\begin{array}{r} \textcircled{1} \ 0 \ 0 \ 0 \\ + \textcircled{0} \ 1 \ 1 \ 1 \\ \hline \textcircled{1} \ 1 \ 1 \ 1 \end{array}$
overflow	overflow	no overflow
(a)	(b)	(c)

If the numbers' sign bits have the same value, which differs from the result's sign bit, overflow has occurred.



Detecting Overflow: Method 2

- Even simpler method: Detect difference between carry-in to sign bit and carry-out from sign bit
- Yields simpler circuit: $\text{overflow} = c3 \text{ xor } c4$

1 1 1 0 1 1 1 + 0 0 0 1 ----- 0 1 0 0 0 overflow (a)	0 0 0 1 1 1 1 + 1 0 0 0 ----- 1 0 1 1 1 overflow (b)	0 0 0 1 0 0 0 + 0 1 1 1 ----- 0 1 1 1 1 no overflow (c)
--	--	---

If the carry into the sign bit column differs from the carry out of that column, overflow has occurred.



Arithmetic-Logic Unit: ALU

- **ALU**: Component that can perform any of various arithmetic (add, subtract, increment, etc.) and logic (AND, OR, etc.) operations, based on control inputs
- Motivation:
 - Suppose want multi-function calculator that not only adds and subtracts, but also increments, ANDs, ORs, XORs, etc.

TABLE 4.2 Desired calculator operations

Inputs			Operation	Sample output if A=00001111, B=00000101
x	y	z		
0	0	0	$S = A + B$	S=00010100
0	0	1	$S = A - B$	S=00001010
0	1	0	$S = A + 1$	S=00010000
0	1	1	$S = A$	S=00001111
1	0	0	$S = A \text{ AND } B$ (bitwise AND)	S=00000101
1	0	1	$S = A \text{ OR } B$ (bitwise OR)	S=00001111
1	1	0	$S = A \text{ XOR } B$ (bitwise XOR)	S=00001010
1	1	1	$S = \text{NOT } A$ (bitwise complement)	S=11110000

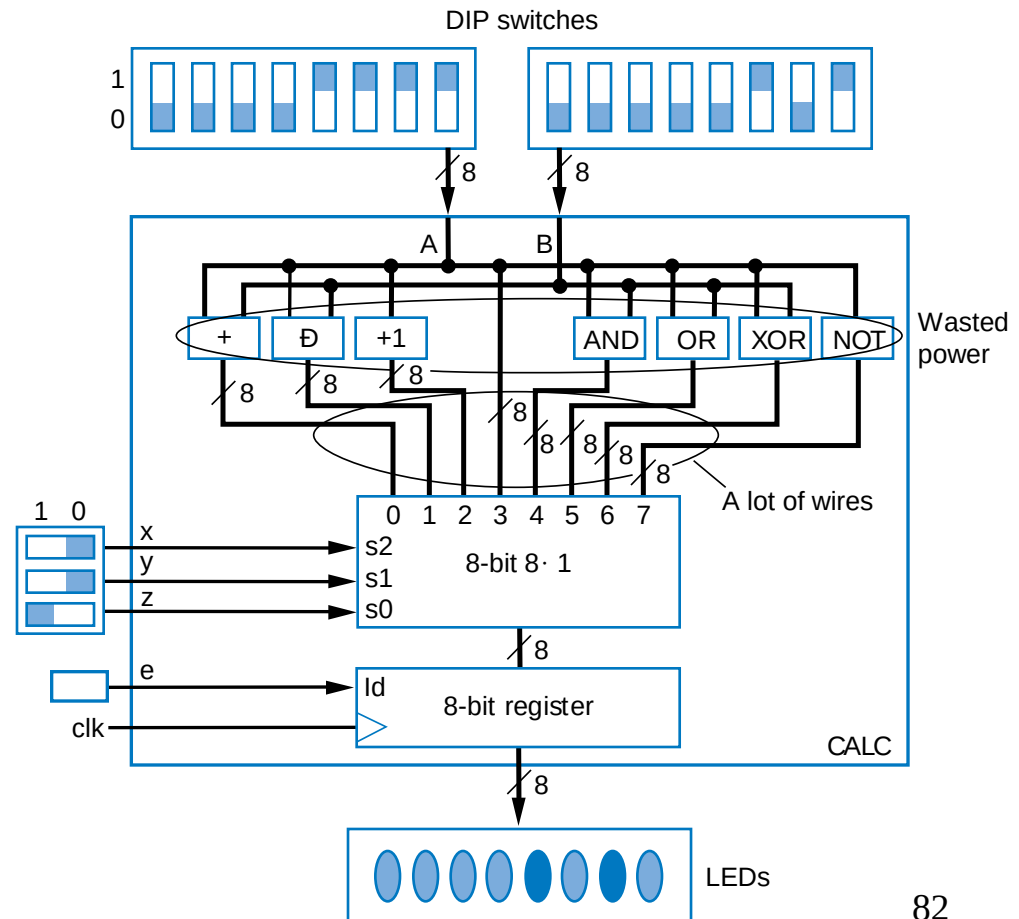


Multifunction Calculator without an ALU

- Can build multifunction calculator using separate components for each operation, and muxes
 - But too many wires, and wasted power computing all those operations when at any time you only use

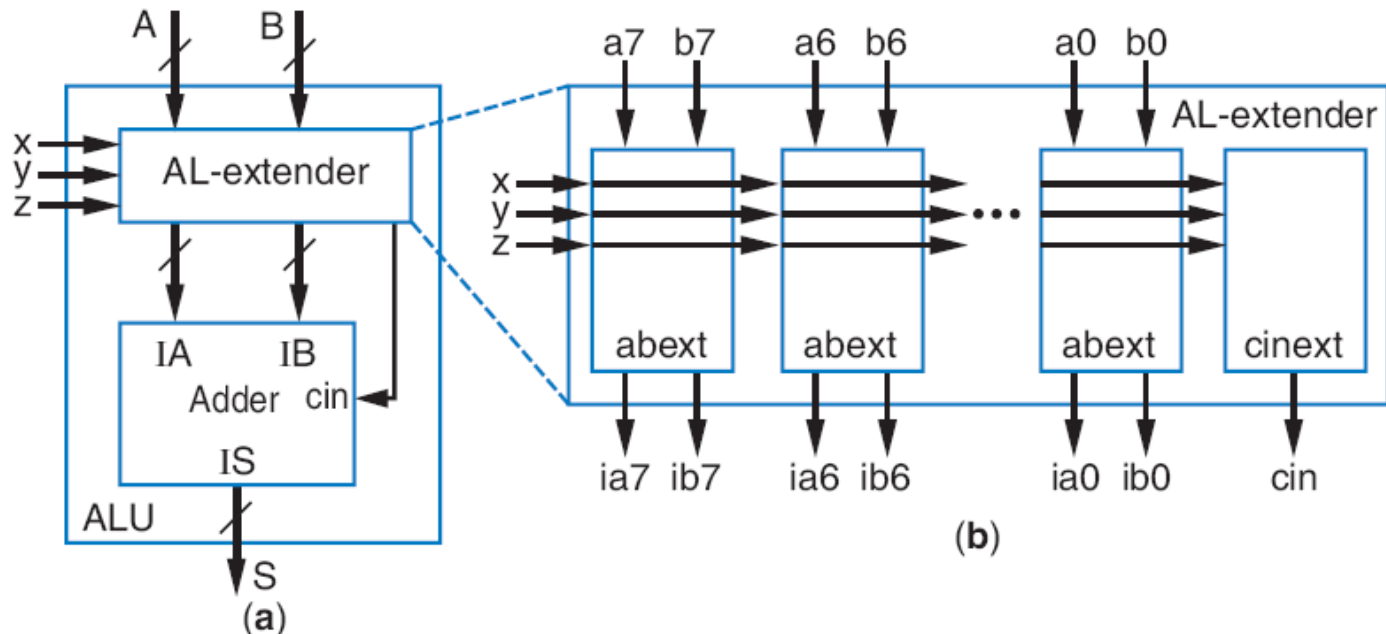
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0	1	1	$S = A$	S=00001111
1	0	0	$S = A \text{ AND } B$ (bitwise AND)	S=00000101
1	0	1	$S = A \text{ OR } B$ (bitwise OR)	S=00001111
1	1	0	$S = A \text{ XOR } B$ (bitwise XOR)	S=00001010
1	1	1	$S = \text{NOT } A$ (bitwise complement)	S=11110000



ALU

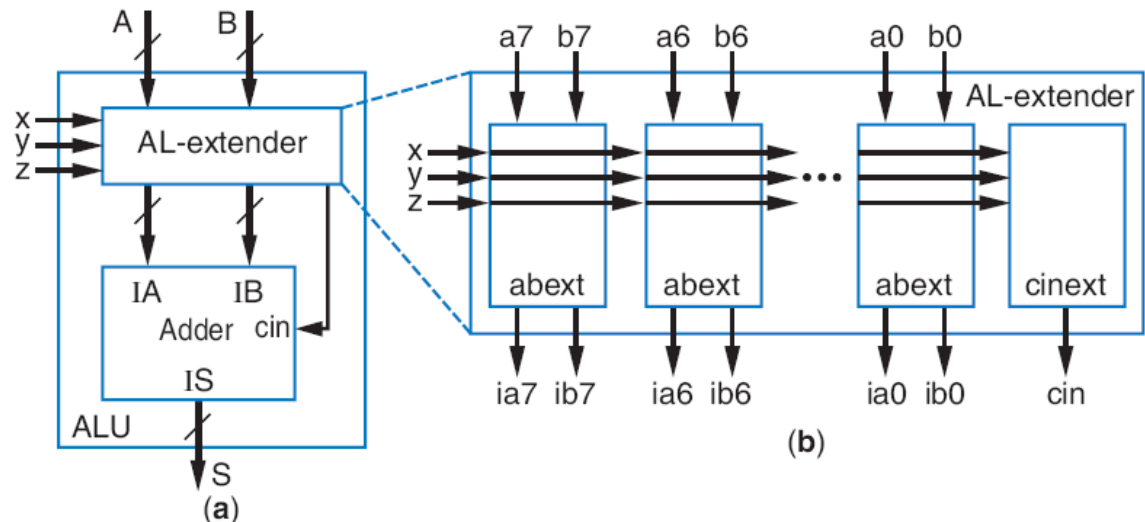
- More efficient design uses ALU
 - ALU design not just separate components multiplexed (same problem as previous slide!),
 - Instead, ALU design uses single adder, plus logic in front of adder's A and B inputs
 - Logic in front is called an arithmetic-logic extender
 - Extender modifies the A and B inputs such that desired operation will appear at output of the adder



Arithmetic-Logic Extender in Front of ALU

TABLE 4.2 Desired calculator operations

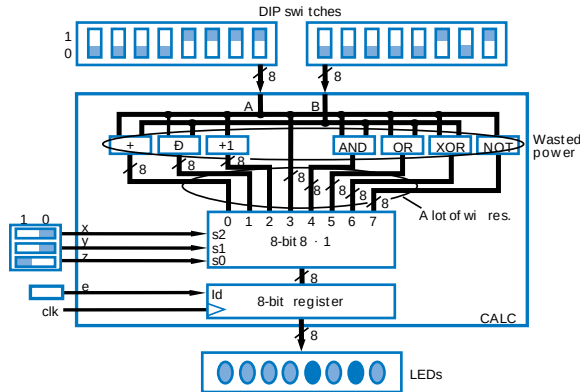
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0	1	1	$S = A$	S=00001111
1	0	0	$S = A \text{ AND } B$ (bitwise AND)	S=00000101
1	0	1	$S = A \text{ OR } B$ (bitwise OR)	S=00001111
1	1	0	$S = A \text{ XOR } B$ (bitwise XOR)	S=00001010
1	1	1	$S = \text{NOT } A$ (bitwise complement)	S=11110000



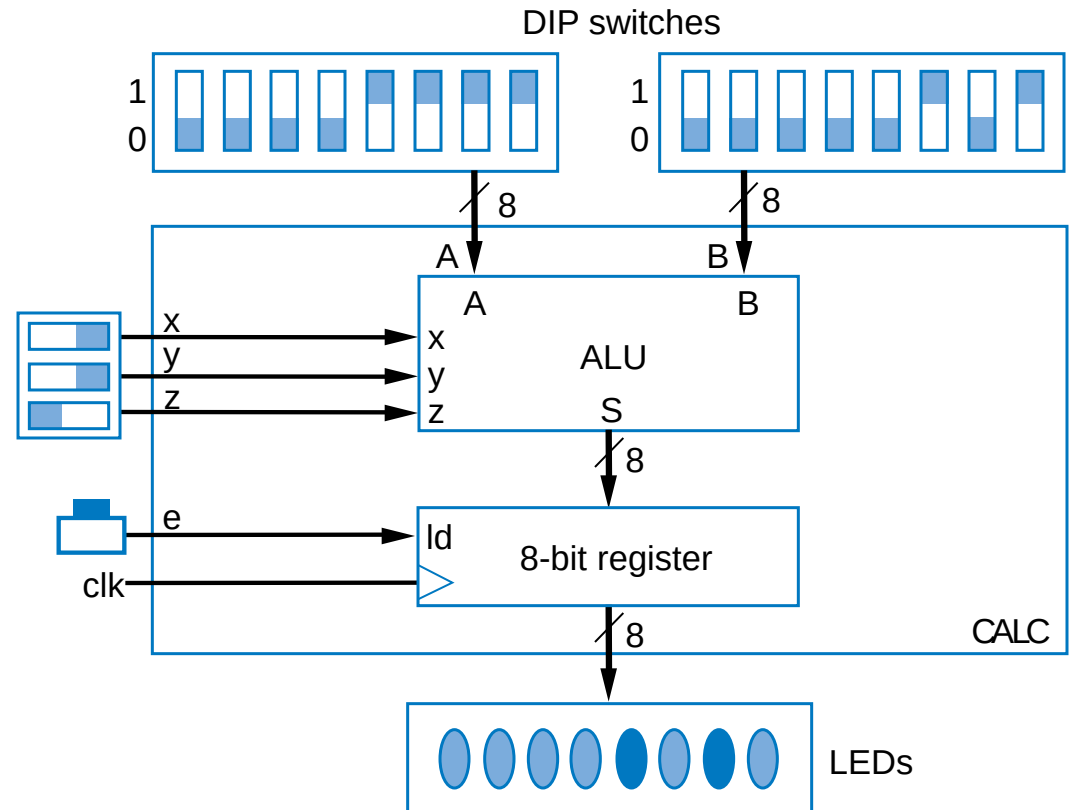
- $xyz=000$: Want $S=A+B$ – just pass a to ia, b to ib, and set $cin=0$
- $xyz=001$: Want $S=A-B$ – pass a to ia, b' to ib, and set $cin=1$
- $xyz=010$: Want $S=A+1$ – pass a to ia, set $ib=0$, and set $cin=1$
- $xyz=011$: Want $S=A$ – pass a to ia, set $ib=0$, and set $cin=0$
- $xyz=1000$: Want $S=A \text{ AND } B$ – set $ia=a*b$, $b=0$, and $cin=0$
- others: likewise
- Based on above, create logic for $ia(x,y,z,a,b)$ and $ib(x,y,z,a,b)$ for each abext, and create logic for $cin(x,y,z)$, to complete design of the AL-extender component



ALU Example: Multifunction Calculator



- Design using ALU is elegant and efficient
 - No mass of wires
 - No big waste of power

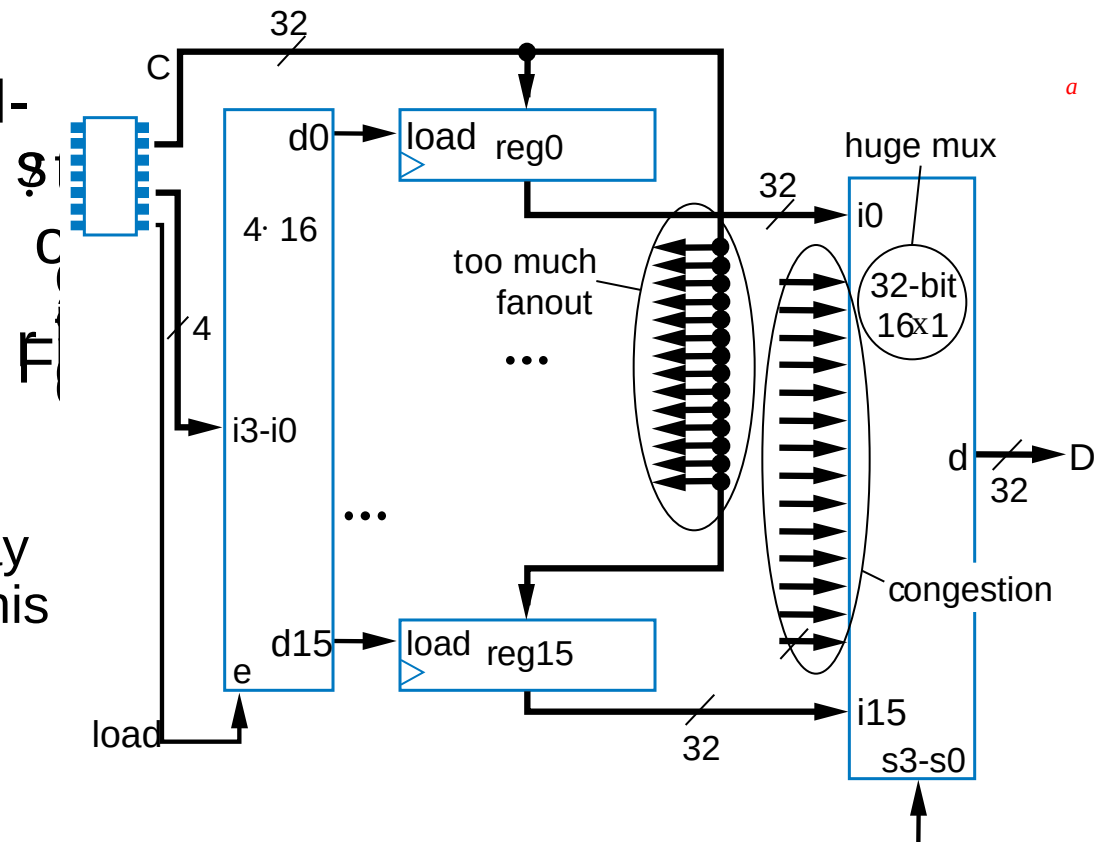


Register Files

- ***MxN register file***

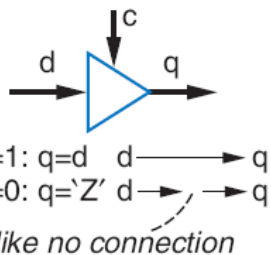
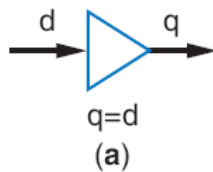
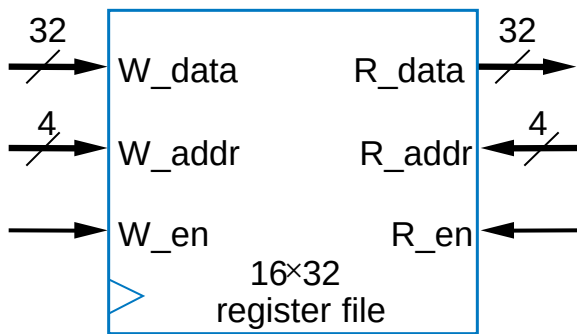
component provides efficient access to M N -bit-wide registers

- If we have many registers but only need access one or two at a time, a register file is more efficient
- Ex: Above-mirror display (earlier example), but this time having 16 32-bit registers
 - Too many wires, and big mux is too slow

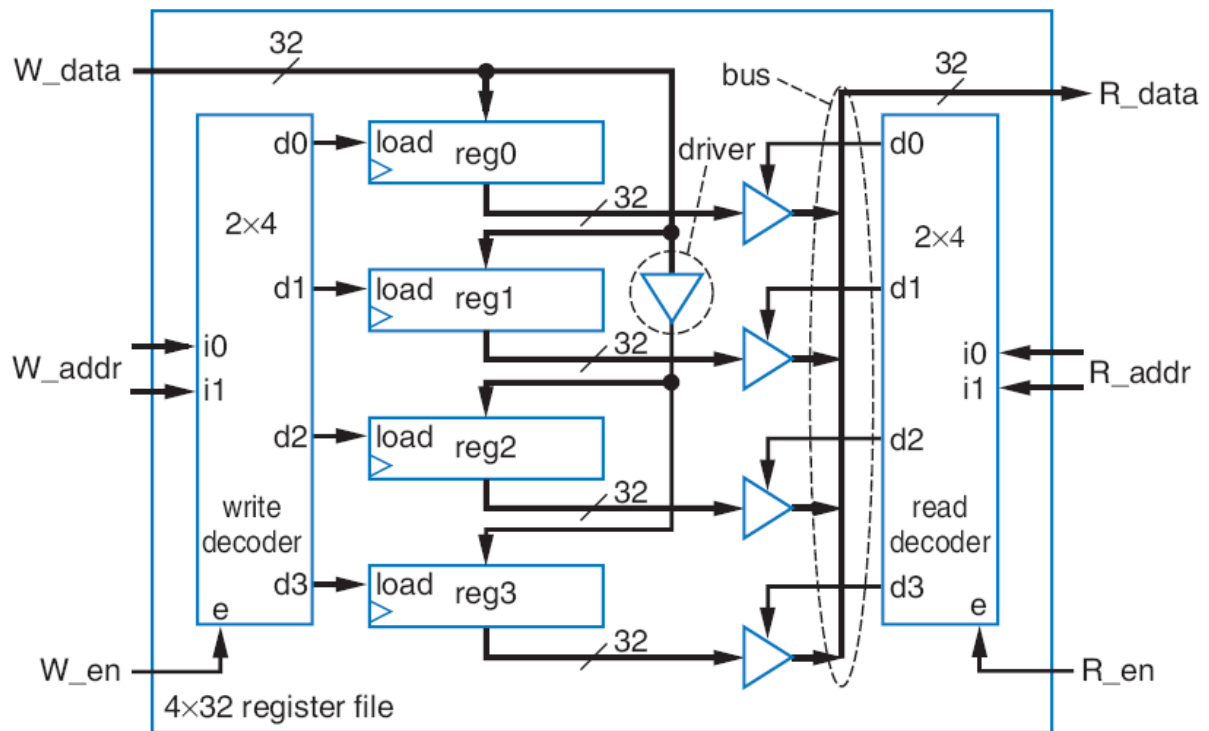


Register File

- Instead, want component that has one data input and one data output, and allows us to specify which internal register to write and which to read

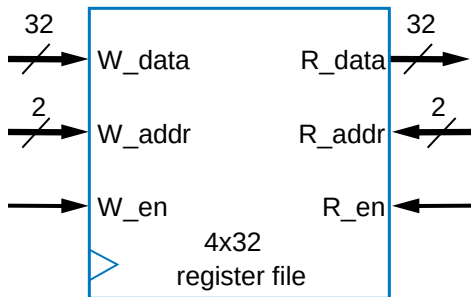
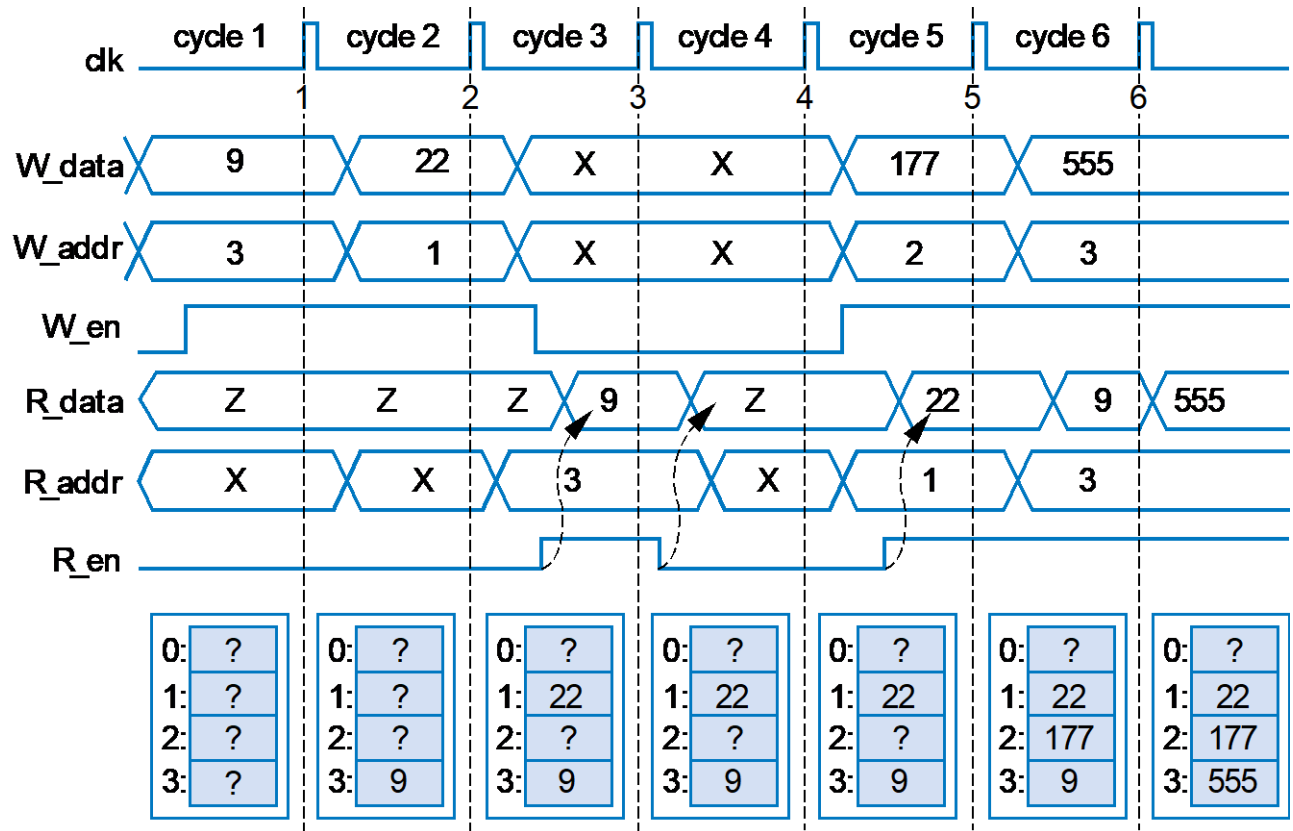


α



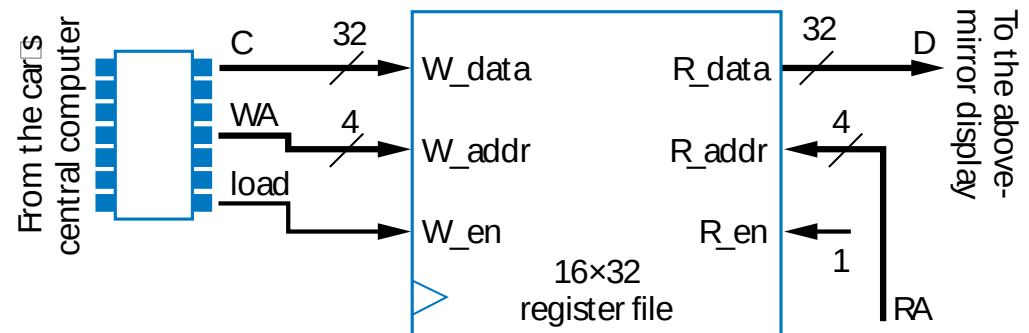
Register File Timing Diagram

- Can write one register and read one register each clock cycle
 - May be same register



Register-File Example: Above-Mirror Display

- 16 32-bit registers that can be written by car's computer, and displayed
 - Use 16x32 register file
 - Simple, elegant design
- Register file hides complexity internally
 - And because only one register needs to be written and/or read at a time, internal design is simple



Chapter Summary

- Need datapath components to store and operate on multibit data
 - Also known as register-transfer-level (RTL) components
- Components introduced
 - Registers
 - Shifters
 - Adders
 - Comparators
 - Counters
 - Multipliers
 - Subtractors
 - Arithmetic-Logic Units
 - Register Files
- Next, we'll combine knowledge of combinational logic design, sequential logic design, and datapath components, to build digital circuits that can perform general and powerful computations

